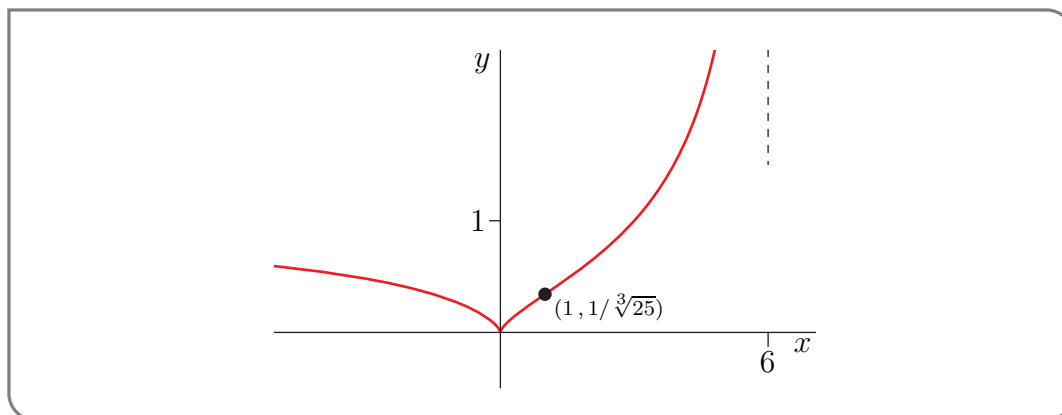
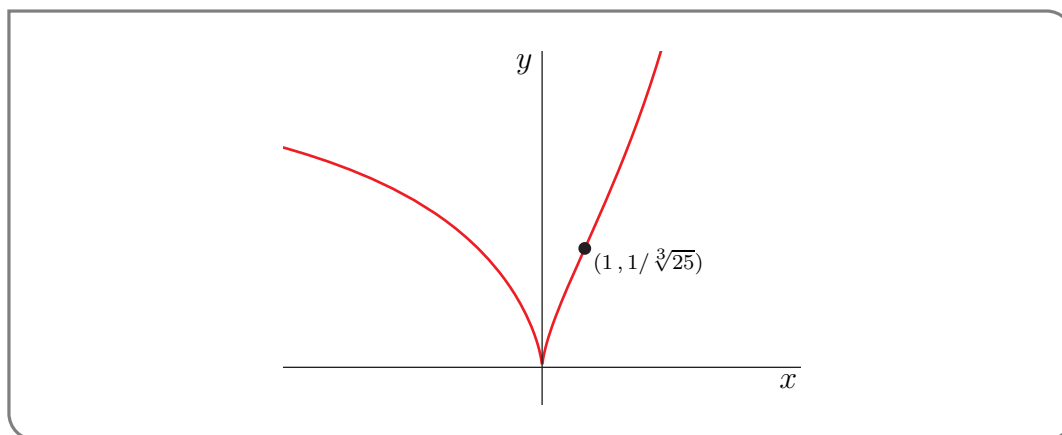


It is hard to see the inflection point at $x = 1$, $y = f(1) = \frac{1}{\sqrt[3]{25}}$ in the above sketch. So here is a blow up of the part of the sketch around $x = 1$.



And if we zoom in even more we have



Example 3.6.16

3.7 ▲ L'Hôpital's Rule and Indeterminate Forms

Let us return to limits (Chapter [1](#)) and see how we can use derivatives to simplify certain families of limits called indeterminate forms. We know, from Theorem [1.4.2](#) on the arithmetic of limits, that if

$$\lim_{x \rightarrow a} f(x) = F$$

$$\lim_{x \rightarrow a} g(x) = G$$

and $G \neq 0$, then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{F}{G}$$

The requirement that $G \neq 0$ is critical — we explored this in Example [1.4.6](#). Please reread that example.

Of course⁶⁴ it is not surprising that if $F \neq 0$ and $G = 0$, then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = DNE$$

and if $F = 0$ but $G \neq 0$ then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = 0$$

However when both $F, G = 0$ then, as we saw in Example 1.4.6, almost anything can happen

$f(x) = x$	$g(x) = x^2$	$\lim_{x \rightarrow 0} \frac{x}{x^2} = \lim_{x \rightarrow 0} \frac{1}{x} = DNE$
$f(x) = x^2$	$g(x) = x$	$\lim_{x \rightarrow 0} \frac{x^2}{x} = \lim_{x \rightarrow 0} x = 0$
$f(x) = x$	$g(x) = x$	$\lim_{x \rightarrow 0} \frac{x}{x} = \lim_{x \rightarrow 0} 1 = 1$
$f(x) = 7x^2$	$g(x) = 3x^2$	$\lim_{x \rightarrow 0} \frac{7x^2}{3x^2} = \lim_{x \rightarrow 0} \frac{7}{3} = \frac{7}{3}$

Indeed after exploring Example 1.4.11 and 1.4.13 we gave ourselves the rule of thumb that if we found $0/0$, then there must be something that cancels.

Because the limit that results from these $0/0$ situations is not immediately obvious, but also leads to some interesting mathematics, we should give it a name.

Definition 3.7.1 (First indeterminate forms).

Let $a \in \mathbb{R}$ and let $f(x)$ and $g(x)$ be functions. If

$$\lim_{x \rightarrow a} f(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = 0$$

then the limit

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$$

is called a $0/0$ indeterminate form.

There are quite a number of mathematical tools for evaluating such indeterminate forms — Taylor series for example. A simpler method, which works in quite a few cases, is L'Hôpital's rule⁶⁵.

⁶⁴ Now it is not so surprising, but perhaps back when we started limits, this was not so obvious.

⁶⁵ Named for the 17th century mathematician, Guillaume de l'Hôpital, who published the first textbook on differential calculus. The eponymous rule appears in that text, but is believed to have been developed by Johann Bernoulli. The book was the source of some controversy since it contained many results by Bernoulli, which l'Hôpital acknowledged in the preface, but Bernoulli felt that l'Hôpital got undue credit.

Theorem 3.7.2 (L'Hôpital's Rule).

Let $a \in \mathbb{R}$ and assume that

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0$$

Then

(a) if $f'(a)$ and $g'(a)$ exist and $g'(a) \neq 0$, then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{f'(a)}{g'(a)},$$

(b) while, if $f'(x)$ and $g'(x)$ exist, with $g'(x)$ nonzero, on an open interval that contains a , except possibly at a itself, and if the limit

$$\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} \text{ exists or is } +\infty \text{ or is } -\infty$$

then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

Proof. We only give the proof for part (a). The proof of part (b) is not very difficult, but uses the Generalised Mean-Value Theorem (Theorem 3.4.38), which is optional and most readers have not seen it.

- First note that we must have $f(a) = g(a) = 0$. To see this note that since derivative $f'(a)$ exists, we know that the limit

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \text{ exists}$$

Since we know that the denominator goes to zero, we must also have that the numerator goes to zero (otherwise the limit would be undefined). Hence we must have

$$\lim_{x \rightarrow a} (f(x) - f(a)) = \left(\lim_{x \rightarrow a} f(x) \right) - f(a) = 0$$

We are told that $\lim_{x \rightarrow a} f(x) = 0$ so we must have $f(a) = 0$. Similarly we know that $g(a) = 0$.

Note that around that time l'Hôpital's name was commonly spelled l'Hospital, but the spelling of silent s in French was changed subsequently; many texts spell his name l'Hospital. If you find yourself in Paris, you can hunt along Boulevard de l'Hôpital for older street signs carved into the sides of buildings which spell it "l'Hospital" — though arguably there are better things to do there.

- Now consider the indeterminate form

$$\begin{aligned}
 \lim_{x \rightarrow a} \frac{f(x)}{g(x)} &= \lim_{x \rightarrow a} \frac{f(x) - 0}{g(x) - 0} && \text{use } 0 = f(a) = g(a) \\
 &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{g(x) - g(a)} && \text{multiply by } 1 = \frac{(x-a)^{-1}}{(x-a)^{-1}} \\
 &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{g(x) - g(a)} \cdot \frac{(x-a)^{-1}}{(x-a)^{-1}} && \text{rearrange} \\
 &= \lim_{x \rightarrow a} \left[\frac{\frac{f(x) - f(a)}{x-a}}{\frac{g(x) - g(a)}{x-a}} \right] && \text{use arithmetic of limits} \\
 &= \frac{\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x-a}}{\lim_{x \rightarrow a} \frac{g(x) - g(a)}{x-a}} = \frac{f'(a)}{g'(a)}
 \end{aligned}$$

We can justify this step and apply Theorem 1.4.2, since the limits in the numerator and denominator exist, because they are just $f'(a)$ and $g'(a)$.

□

►►► Optional — Proof of Part (b) of l'Hôpital's Rule

To prove part (b) we must work around the possibility that $f'(a)$ and $g'(a)$ do not exist or that $f'(x)$ and $g'(x)$ are not continuous at $x = a$. To do this, we make use of the Generalised Mean-Value Theorem (Theorem 3.4.38) that was used to prove Equation (3.4.33). We recommend you review the GMVT before proceeding.

For simplicity we consider the limit

$$\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)}$$

By assumption, we know that

$$\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^+} g(x) = 0$$

For simplicity, we also assume that $f(a) = g(a) = 0$. This allows us to write

$$\frac{f(x)}{g(x)} = \frac{f(x) - f(a)}{g(x) - g(a)}$$

which is the right form for an application of the GMVT.

By assumption $f'(x)$ and $g'(x)$ exist, with $g'(x)$ nonzero, in some open interval around a , except possibly at a itself. So we know that they exist, with $g'(x) \neq 0$, in some interval $(a, b]$ with $b > a$. Then the GMVT (Theorem 3.4.38) tells us that for $x \in (a, b]$

$$\frac{f(x)}{g(x)} = \frac{f(x) - f(a)}{g(x) - g(a)} = \frac{f'(c)}{g'(c)}$$

where $c \in (a, x)$. As we take the limit as $x \rightarrow a$, we also have that $c \rightarrow a$, and so

$$\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a^+} \frac{f'(c)}{g'(c)} = \lim_{c \rightarrow a^+} \frac{f'(c)}{g'(c)}$$

as required.

3.7.1 ► Standard Examples

Here are some simple examples using L'Hôpital's rule.

Example 3.7.3

Consider the limit

$$\lim_{x \rightarrow 0} \frac{\sin x}{x}$$

- Notice that

$$\lim_{x \rightarrow 0} \sin x = 0$$

$$\lim_{x \rightarrow 0} x = 0$$

so this is a $0/0$ indeterminate form, and suggests we try l'Hôpital's rule.

- To apply the rule we must first check the limits of the derivatives.

$$f(x) = \sin x$$

$$f'(x) = \cos x$$

and

$$f'(0) = 1$$

$$g(x) = x$$

$$g'(x) = 1$$

and

$$g'(0) = 1$$

- So by l'Hôpital's rule

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = \frac{f'(0)}{g'(0)} = \frac{1}{1} = 1.$$

Example 3.7.3

Example 3.7.4

Consider the limit

$$\lim_{x \rightarrow 0} \frac{\sin(x)}{\sin(2x)}$$

- First check

$$\lim_{x \rightarrow 0} \sin 2x = 0$$

$$\lim_{x \rightarrow 0} \sin x = 0$$

so we again have a $0/0$ indeterminate form.

- Set $f(x) = \sin x$ and $g(x) = \sin 2x$, then

$$\begin{aligned} f'(x) &= \cos x & f'(0) &= 1 \\ g'(x) &= 2 \cos 2x & g'(0) &= 2 \end{aligned}$$

- And by l'Hôpital's rule

$$\lim_{x \rightarrow 0} \frac{\sin x}{\sin 2x} = \frac{f'(0)}{g'(0)} = \frac{1}{2}.$$

Example 3.7.4

Example 3.7.5

Let $q > 1$ and compute the limit

$$\lim_{x \rightarrow 0} \frac{q^x - 1}{x}$$

This limit arose in our discussion of exponential functions in Section 2.7.

- First check

$$\begin{aligned} \lim_{x \rightarrow 0} (q^x - 1) &= 1 - 1 = 0 \\ \lim_{x \rightarrow 0} x &= 0 \end{aligned}$$

so we have a $0/0$ indeterminate form.

- Set $f(x) = q^x - 1$ and $g(x) = x$, then (maybe after a quick review of Section 2.7)

$$\begin{aligned} f'(x) &= \frac{d}{dx} (q^x - 1) = q^x \cdot \log q & f'(0) &= \log q \\ g'(x) &= 1 & g'(0) &= 1 \end{aligned}$$

- And by l'Hôpital's rule⁶⁶

$$\lim_{h \rightarrow 0} \frac{q^h - 1}{h} = \log q.$$

⁶⁶ While it might not be immediately obvious, this example relies on circular reasoning. In order to apply l'Hôpital's rule, we need to compute the derivative of q^x . However in order to compute that limit (see Section 2.7) we needed to evaluate this limit.

A more obvious example of this sort of circular reasoning can be seen if we use l'Hôpital's rule to compute the derivative of $f(x) = x^n$ at $x = a$ using the limit

$$f'(a) = \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = \lim_{x \rightarrow a} \frac{nx^{n-1} - 0}{1 - 0} = na^{n-1}.$$

We have used the result $\frac{d}{dx} x^n = nx^{n-1}$ to prove itself!

Example 3.7.5

In this example, we shall apply L'Hôpital's rule twice before getting the answer.

Example 3.7.6

Compute the limit

$$\lim_{x \rightarrow 0} \frac{\sin(x^2)}{1 - \cos x}$$

- Again we should check

$$\begin{aligned}\lim_{x \rightarrow 0} \sin(x^2) &= \sin 0 = 0 \\ \lim_{x \rightarrow 0} (1 - \cos x) &= 1 - \cos 0 = 0\end{aligned}$$

and we have a $0/0$ indeterminate form.

- Let $f(x) = \sin(x^2)$ and $g(x) = 1 - \cos x$ then

$$\begin{aligned}f'(x) &= 2x \cos(x^2) & f'(0) &= 0 \\ g'(x) &= \sin x & g'(0) &= 0\end{aligned}$$

So if we try to apply l'Hôpital's rule naively we will get

$$\lim_{x \rightarrow 0} \frac{\sin(x^2)}{1 - \cos x} = \frac{f'(0)}{g'(0)} = \frac{0}{0}.$$

which is another $0/0$ indeterminate form.

- It appears that we are stuck until we remember that l'Hôpital's rule (as stated in Theorem [3.7.2](#)) has a part (b) — now is a good time to reread it.
- It says that

$$\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = \lim_{x \rightarrow 0} \frac{f'(x)}{g'(x)}$$

provided this second limit exists. In our case this requires us to compute

$$\lim_{x \rightarrow 0} \frac{2x \cos(x^2)}{\sin(x)}$$

which we can do using l'Hôpital's rule again. Now

$$\begin{aligned}h(x) &= 2x \cos(x^2) & h'(x) &= 2 \cos(x^2) - 4x^2 \sin(x^2) & h'(0) &= 2 \\ \ell(x) &= \sin(x) & \ell'(x) &= \cos(x) & \ell'(0) &= 1\end{aligned}$$

By l'Hôpital's rule

$$\lim_{x \rightarrow 0} \frac{2x \cos(x^2)}{\sin(x)} = \frac{h'(0)}{\ell'(0)} = 2$$

- Thus our original limit is

$$\lim_{x \rightarrow 0} \frac{\sin(x^2)}{1 - \cos x} = \lim_{x \rightarrow 0} \frac{2x \cos(x^2)}{\sin(x)} = 2.$$

- We can succinctly summarise the two applications of L'Hôpital's rule in this example by

$$\lim_{x \rightarrow 0} \underbrace{\frac{\sin(x^2)}{1 - \cos x}}_{\substack{\text{num} \rightarrow 0 \\ \text{den} \rightarrow 0}} = \lim_{x \rightarrow 0} \underbrace{\frac{2x \cos(x^2)}{\sin x}}_{\substack{\text{num} \rightarrow 0 \\ \text{den} \rightarrow 0}} = \lim_{x \rightarrow 0} \underbrace{\frac{2 \cos(x^2) - 4x^2 \sin(x^2)}{\cos x}}_{\substack{\text{num} \rightarrow 2 \\ \text{den} \rightarrow 1}} = 2$$

Here “num” and “den” are used as abbreviations of “numerator” and “denominator” respectively.

Example 3.7.6

One must be careful to ensure that the hypotheses of l'Hôpital's rule are satisfied before applying it. The following “warnings” show the sorts of things that can go wrong.

Warning 3.7.7 (Denominator limit nonzero).

If

$$\lim_{x \rightarrow a} f(x) = 0 \quad \text{but} \quad \lim_{x \rightarrow a} g(x) \neq 0$$

then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} \quad \text{need not be the same as} \quad \frac{f'(a)}{g'(a)} \text{ or } \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}.$$

Here is an example. Take

$$a = 0 \quad f(x) = 3x \quad g(x) = 4 + 5x$$

Then

$$\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = \lim_{x \rightarrow 0} \frac{3x}{4 + 5x} = \frac{3 \times 0}{4 + 5 \times 0} = 0$$

$$\lim_{x \rightarrow 0} \frac{f'(x)}{g'(x)} = \frac{f'(0)}{g'(0)} = \frac{3}{5}$$

Warning 3.7.8 (Numerator limit nonzero).

If

$$\lim_{x \rightarrow a} g(x) = 0 \quad \text{but} \quad \lim_{x \rightarrow a} f(x) \neq 0$$

then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} \quad \text{need not be the same as} \quad \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}.$$

Here is an example. Take

$$a = 0 \quad f(x) = 4 + 5x \quad g(x) = 3x$$

Then

$$\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = \lim_{x \rightarrow 0} \frac{4 + 5x}{3x} = \text{DNE}$$

$$\lim_{x \rightarrow 0} \frac{f'(x)}{g'(x)} = \lim_{x \rightarrow 0} \frac{5}{3} = \frac{5}{3}$$

This next one is more subtle; the limits of the original numerator and denominator functions both go to zero, but the limit of the ratio their derivatives does not exist.

Warning 3.7.9 (Limit of ratio of derivatives DNE).

If

$$\lim_{x \rightarrow a} f(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = 0$$

but

$$\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} \text{ does not exist}$$

then it is still possible that

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} \text{ exists.}$$

Here is an example. Take

$$a = 0 \quad f(x) = x^2 \sin \frac{1}{x} \quad g(x) = x$$

Then (with an application of the squeeze theorem)

$$\lim_{x \rightarrow 0} f(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow 0} g(x) = 0.$$

If we attempt to apply l'Hôpital's rule then we have $g'(x) = 1$ and

$$f'(x) = 2x \sin \frac{1}{x} - \cos \frac{1}{x}$$

and we then try to compute the limit

$$\lim_{x \rightarrow 0} \frac{f'(x)}{g'(x)} = \lim_{x \rightarrow 0} \left(2x \sin \frac{1}{x} - \cos \frac{1}{x} \right)$$

However, this limit does not exist. The first term converges to 0 (by the squeeze theorem), but the second term $\cos(1/x)$ just oscillates wildly between ± 1 . All we can conclude from this is

Since the limit of the ratio of derivatives does not exist, we cannot apply l'Hôpital's rule.

Instead we should go back to the original limit and apply the squeeze theorem:

$$\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = \lim_{x \rightarrow 0} \frac{x^2 \sin \frac{1}{x}}{x} = \lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0,$$

since $|x \sin(1/x)| < |x|$ and $|x| \rightarrow 0$ as $x \rightarrow 0$.

It is also easy to construct an example in which the limits of numerator and denominator are both zero, but the limit of the ratio and the limit of the ratio of the derivatives do not exist. A slight change of the previous example shows that it is possible that

$$\lim_{x \rightarrow a} f(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = 0$$

but neither of the limits

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} \quad \text{or} \quad \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

exist. Take

$$a = 0 \quad f(x) = x \sin \frac{1}{x} \quad g(x) = x$$

Then (with a quick application of the squeeze theorem)

$$\lim_{x \rightarrow 0} f(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow 0} g(x) = 0.$$

However,

$$\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = \lim_{x \rightarrow 0} \frac{x \sin \frac{1}{x}}{x} = \lim_{x \rightarrow 0} \sin \frac{1}{x}$$

does not exist. And similarly

$$\lim_{x \rightarrow 0} \frac{f'(x)}{g'(x)} = \lim_{x \rightarrow 0} \frac{\sin \frac{1}{x} - \frac{1}{x} \cos \frac{1}{x}}{x^2}$$

does not exist.

3.7.2 ► Variations

Theorem [3.7.2](#) is the basic form of L'Hôpital's rule, but there are also many variations. Here are a bunch of them.

- (a) L'Hôpital's rule also applies when the limit of $x \rightarrow a$ is replaced by $\lim_{x \rightarrow a+}$ or by $\lim_{x \rightarrow a-}$ or by $\lim_{x \rightarrow +\infty}$ or by $\lim_{x \rightarrow -\infty}$.

We can justify adapting the rule to the limits to $\pm\infty$ via the following reasoning

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} &= \lim_{y \rightarrow 0^+} \frac{f(1/y)}{g(1/y)} && \text{substitute } x = 1/y \\ &= \lim_{y \rightarrow 0^+} \frac{-\frac{1}{y^2} f'(1/y)}{-\frac{1}{y^2} g'(1/y)} \end{aligned}$$

where we have used l'Hôpital's rule (assuming this limit exists) and the fact that $\frac{d}{dy}f(1/y) = -\frac{1}{y^2}f'(1/y)$ (and similarly for g). Cleaning this up and substituting $y = 1/x$ gives the required result:

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{y \rightarrow 0^+} \frac{f'(1/y)}{g'(1/y)} = \lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)}.$$

Example 3.7.10

Consider the limit

$$\lim_{x \rightarrow \infty} \frac{\arctan x - \frac{\pi}{2}}{1/x}$$

Both numerator and denominator go to 0 as $x \rightarrow \infty$, so this is an $0/0$ indeterminate form. We find

$$\lim_{x \rightarrow +\infty} \underbrace{\frac{\arctan x - \frac{\pi}{2}}{\frac{1}{x}}}_{\substack{\text{num} \rightarrow 0 \\ \text{den} \rightarrow 0}} = \lim_{x \rightarrow +\infty} \frac{\frac{1}{1+x^2}}{-\frac{1}{x^2}} = - \lim_{x \rightarrow +\infty} \underbrace{\frac{1}{1 + \frac{1}{x^2}}}_{\substack{\text{num} \rightarrow 1 \\ \text{den} \rightarrow 1}} = -1$$

We have applied L'Hôpital's rule with

$$\begin{aligned} f(x) &= \arctan x - \frac{\pi}{2} & g(x) &= \frac{1}{x} \\ f'(x) &= \frac{1}{1+x^2} & g'(x) &= -\frac{1}{x^2} \end{aligned}$$

Example 3.7.10

- (b) $\frac{\infty}{\infty}$ indeterminate form: L'Hôpital's rule also applies when $\lim_{x \rightarrow a} f(x) = 0, \lim_{x \rightarrow a} g(x) = 0$ is replaced by $\lim_{x \rightarrow a} f(x) = \pm\infty, \lim_{x \rightarrow a} g(x) = \pm\infty$.

Example 3.7.11

Consider the limit

$$\lim_{x \rightarrow \infty} \frac{\log x}{x}$$

The numerator and denominator both blow up towards infinity so this is an ∞/∞ indeterminate form. An application of l'Hôpital's rule gives

$$\begin{aligned} \lim_{x \rightarrow \infty} \underbrace{\frac{\log x}{x}}_{\substack{\text{num} \rightarrow \infty \\ \text{den} \rightarrow \infty}} &= \lim_{x \rightarrow \infty} \frac{1/x}{1} \\ &= \lim_{x \rightarrow \infty} \frac{1}{x} = 0 \end{aligned}$$

Example 3.7.11

Example 3.7.12

Consider the limit

$$\lim_{x \rightarrow \infty} \frac{5x^2 + 3x - 3}{x^2 + 1}$$

Then by two applications of l'Hôpital's rule we get

$$\lim_{x \rightarrow \infty} \underbrace{\frac{5x^2 + 3x - 3}{x^2 + 1}}_{\substack{\text{num} \rightarrow \infty \\ \text{den} \rightarrow \infty}} = \lim_{x \rightarrow \infty} \underbrace{\frac{10x + 3}{2x}}_{\substack{\text{num} \rightarrow \infty \\ \text{den} \rightarrow \infty}} = \lim_{x \rightarrow \infty} \frac{10}{2} = 5.$$

Example 3.7.12

Example 3.7.13

Compute the limit

$$\lim_{x \rightarrow 0^+} \frac{\log x}{\tan\left(\frac{\pi}{2} - x\right)}$$

We can compute this using l'Hôpital's rule twice:

$$\begin{aligned} \lim_{x \rightarrow 0^+} \underbrace{\frac{\log x}{\tan\left(\frac{\pi}{2} - x\right)}}_{\substack{\text{num} \rightarrow -\infty \\ \text{den} \rightarrow +\infty}} &= \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{-\sec^2\left(\frac{\pi}{2} - x\right)} = - \lim_{x \rightarrow 0^+} \underbrace{\frac{\cos^2\left(\frac{\pi}{2} - x\right)}{x}}_{\substack{\text{num} \rightarrow 0 \\ \text{den} \rightarrow 0}} \\ &= - \lim_{x \rightarrow 0^+} \underbrace{\frac{2 \cos\left(\frac{\pi}{2} - x\right) \sin\left(\frac{\pi}{2} - x\right)}{1}}_{\substack{\text{num} \rightarrow 0 \\ \text{den} \rightarrow 1}} = 0 \end{aligned}$$

The first application of L'Hôpital's was with

$$\begin{aligned} f(x) &= \log x & g(x) &= \tan\left(\frac{\pi}{2} - x\right) \\ f'(x) &= \frac{1}{x} & g'(x) &= -\sec^2\left(\frac{\pi}{2} - x\right) \end{aligned}$$

and the second time with

$$\begin{aligned} f(x) &= \cos^2\left(\frac{\pi}{2} - x\right) & g(x) &= x \\ f'(x) &= 2 \cos\left(\frac{\pi}{2} - x\right) \left[-\sin\left(\frac{\pi}{2} - x\right)\right](-1) & g'(x) &= 1 \end{aligned}$$

Example 3.7.13

Sometimes things don't quite work out as we would like and l'Hôpital's rule can get stuck in a loop. Remember to think about the problem before you apply any rule.

Example 3.7.14

Consider the limit

$$\lim_{x \rightarrow \infty} \frac{e^x + e^{-x}}{e^x - e^{-x}}$$

Clearly both numerator and denominator go to ∞ , so we have a ∞/∞ indeterminate form. Naively applying l'Hôpital's rule gives

$$\lim_{x \rightarrow \infty} \frac{e^x + e^{-x}}{e^x - e^{-x}} = \lim_{x \rightarrow \infty} \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

which is again a ∞/∞ indeterminate form. So apply l'Hôpital's rule again:

$$\lim_{x \rightarrow \infty} \frac{e^x - e^{-x}}{e^x + e^{-x}} = \lim_{x \rightarrow \infty} \frac{e^x + e^{-x}}{e^x - e^{-x}}$$

which is right back where we started!

The correct approach to such a limit is to apply the methods we learned in Chapter 1 and rewrite

$$\frac{e^x + e^{-x}}{e^x - e^{-x}} = \frac{e^x(1 + e^{-2x})}{e^x(1 - e^{-2x})} = \frac{1 + e^{-2x}}{1 - e^{-2x}}$$

and then take the limit.

A similar sort of l'Hôpital-rule-loop will occur if you naively apply l'Hôpital's rule to the limit

$$\lim_{x \rightarrow \infty} \frac{\sqrt{4x^2 + 1}}{5x - 1}$$

which appeared in Example 1.5.6.

Example 3.7.14

►►► Optional — Proof of l'Hôpital's Rule for ∞/∞ forms

We can justify this generalisation of l'Hôpital's rule with some careful manipulations. Since the derivatives f', g' exist in some interval around a , we know that f, g are con-

tinuous in some interval around a ; let x, t be points inside that interval. Now rewrite⁶⁷

$$\begin{aligned}
 \frac{f(x)}{g(x)} &= \frac{f(x)}{g(x)} + \underbrace{\left(\frac{f(t)}{g(x)} - \frac{f(t)}{g(x)} \right)}_{=0} + \underbrace{\left(\frac{f(x) - f(t)}{g(x) - g(t)} - \frac{f(x) - f(t)}{g(x) - g(t)} \right)}_{=0} \\
 &= \underbrace{\frac{f(x) - f(t)}{g(x) - g(t)}}_{\text{ready for GMVT}} + \frac{f(t)}{g(x)} + \underbrace{\left(\frac{f(x)}{g(x)} - \frac{f(t)}{g(x)} - \frac{f(x) - f(t)}{g(x) - g(t)} \right)}_{\text{we can clean it up}} \\
 &= \frac{f(x) - f(t)}{g(x) - g(t)} + \frac{f(t)}{g(x)} + \left(\frac{f(x) - f(t)}{g(x)} - \frac{f(x) - f(t)}{g(x) - g(t)} \right) \\
 &= \frac{f(x) - f(t)}{g(x) - g(t)} + \frac{f(t)}{g(x)} + \left(\frac{1}{g(x)} - \frac{1}{g(x) - g(t)} \right) \cdot (f(x) - f(t)) \\
 &= \frac{f(x) - f(t)}{g(x) - g(t)} + \frac{f(t)}{g(x)} + \left(\frac{g(x) - g(t) - g(x)}{g(x)(g(x) - g(t))} \right) \cdot (f(x) - f(t)) \\
 &= \underbrace{\frac{f(x) - f(t)}{g(x) - g(t)}}_{\text{ready for GMVT}} + \frac{f(t)}{g(x)} - \frac{g(t)}{g(x)} \cdot \underbrace{\frac{f(x) - f(t)}{g(x) - g(t)}}_{\text{ready for GMVT}}
 \end{aligned}$$

Oof! Now the generalised mean-value theorem (Theorem 3.4.38) tells us there is a c between x and t so that

$$\frac{f(x) - f(t)}{g(x) - g(t)} = \frac{f'(c)}{g'(c)}$$

Now substitute this into the large expression we derived above:

$$\frac{f(x)}{g(x)} = \frac{f'(c)}{g'(c)} + \frac{1}{g(x)} \left(f(t) - \frac{f'(c)}{g'(c)} \cdot g(t) \right)$$

At first glance this does not appear so useful, however if we fix t and take the limit as $x \rightarrow a$, then it becomes

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(c)}{g'(c)} + \lim_{x \rightarrow a} \frac{1}{g(x)} \left(f(t) - \frac{f'(c)}{g'(c)} \cdot g(t) \right)$$

Since $g(x) \rightarrow \infty$ as $x \rightarrow a$, this last term goes to zero

$$= \lim_{x \rightarrow a} \frac{f'(c)}{g'(c)} + 0$$

Now take the limit as $t \rightarrow a$. The left-hand side is unchanged since it is independent of t . The right-hand side, however, does change; the number c is trapped between x and t . Since we have already taken the limit $x \rightarrow a$, so when we take the limit $t \rightarrow a$, we are effectively taking the limit $c \rightarrow a$. Hence

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{c \rightarrow a} \frac{f'(c)}{g'(c)}$$

which is the desired result.

⁶⁷ This is quite a clever argument, but it is not immediately obvious why one rewrites things this way. After the fact it becomes clear that it is done to massage the expression into the form where we can apply the generalised mean-value theorem (Theorem 3.4.38).

(c) $0 \cdot \infty$ indeterminate form: When $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = \infty$. We can use a little algebra to manipulate this into either a $\frac{0}{0}$ or $\frac{\infty}{\infty}$ form:

$$\lim_{x \rightarrow a} \frac{f(x)}{1/g(x)} \qquad \lim_{x \rightarrow a} \frac{g(x)}{1/f(x)}$$

Example 3.7.15

Consider the limit

$$\lim_{x \rightarrow 0^+} x \cdot \log x$$

Here the function $f(x) = x$ goes to zero, while $g(x) = \log x$ goes to $-\infty$. If we rewrite this as the fraction

$$x \cdot \log x = \frac{\log x}{1/x}$$

then the $0 \cdot \infty$ form has become an ∞/∞ form.

The result is then

$$\lim_{x \rightarrow 0^+} \underbrace{x}_{\rightarrow 0} \underbrace{\log x}_{\rightarrow -\infty} = \lim_{x \rightarrow 0^+} \frac{\log x}{\underbrace{\frac{1}{x}}_{\substack{\text{num} \rightarrow -\infty \\ \text{den} \rightarrow \infty}}} = \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{-\frac{1}{x^2}} = - \lim_{x \rightarrow 0^+} x = 0$$

Example 3.7.15

Example 3.7.16

In this example we'll evaluate $\lim_{x \rightarrow +\infty} x^n e^{-x}$, for all natural numbers n . We'll start with $n = 1$ and $n = 2$ and then, using what we have learned from those cases, move on to general n .

$$\lim_{x \rightarrow +\infty} \underbrace{x}_{\rightarrow \infty} \underbrace{e^{-x}}_{\rightarrow 0} = \lim_{x \rightarrow +\infty} \frac{x}{\underbrace{e^x}_{\substack{\text{num} \rightarrow +\infty \\ \text{den} \rightarrow +\infty}}} = \lim_{x \rightarrow +\infty} \frac{1}{\underbrace{e^x}_{\substack{\text{num} \rightarrow 1 \\ \text{den} \rightarrow +\infty}}} = \lim_{x \rightarrow +\infty} e^{-x} = 0$$

Applying l'Hôpital twice,

$$\lim_{x \rightarrow +\infty} \underbrace{x^2}_{\rightarrow \infty} \underbrace{e^{-x}}_{\rightarrow 0} = \lim_{x \rightarrow +\infty} \frac{x^2}{\underbrace{e^x}_{\substack{\text{num} \rightarrow +\infty \\ \text{den} \rightarrow +\infty}}} = \lim_{x \rightarrow +\infty} \frac{2x}{\underbrace{e^x}_{\substack{\text{num} \rightarrow \infty \\ \text{den} \rightarrow +\infty}}} = \lim_{x \rightarrow +\infty} \frac{2}{\underbrace{e^x}_{\substack{\text{num} \rightarrow 2 \\ \text{den} \rightarrow +\infty}}} = \lim_{x \rightarrow +\infty} 2e^{-x} = 0$$

Indeed, for any natural number n , applying l'Hôpital n times gives

$$\begin{aligned}
 \lim_{x \rightarrow +\infty} \underbrace{x^n}_{\rightarrow \infty} \underbrace{e^{-x}}_{\rightarrow 0} &= \lim_{x \rightarrow +\infty} \underbrace{\frac{x^n}{e^x}}_{\substack{\text{num} \rightarrow +\infty \\ \text{den} \rightarrow +\infty}} \\
 &= \lim_{x \rightarrow +\infty} \underbrace{\frac{nx^{n-1}}{e^x}}_{\substack{\text{num} \rightarrow \infty \\ \text{den} \rightarrow +\infty}} \\
 &= \lim_{x \rightarrow +\infty} \underbrace{\frac{n(n-1)x^{n-2}}{e^x}}_{\substack{\text{num} \rightarrow \infty \\ \text{den} \rightarrow +\infty}} \\
 &= \cdots = \lim_{x \rightarrow +\infty} \underbrace{\frac{n!}{e^x}}_{\substack{\text{num} \rightarrow n! \\ \text{den} \rightarrow +\infty}} = 0
 \end{aligned}$$

Example 3.7.16

(d) $\infty - \infty$ indeterminate form: When $\lim_{x \rightarrow a} f(x) = \infty$ and $\lim_{x \rightarrow a} g(x) = \infty$. We rewrite the difference as a fraction using a common denominator

$$f(x) - g(x) = \frac{h(x)}{\ell(x)}$$

which is then a $0/0$ or ∞/∞ form.

Example 3.7.17

Consider the limit

$$\lim_{x \rightarrow \frac{\pi}{2}^-} (\sec x - \tan x)$$

Since the limit of both $\sec x$ and $\tan x$ is $+\infty$ as $x \rightarrow \frac{\pi}{2}^-$, this is an $\infty - \infty$ indeterminate form. However we can rewrite this as

$$\sec x - \tan x = \frac{1}{\cos x} - \frac{\sin x}{\cos x} = \frac{1 - \sin x}{\cos x}$$

which is then a $0/0$ indeterminate form. This then gives

$$\begin{aligned}
 \lim_{x \rightarrow \frac{\pi}{2}^-} \left(\underbrace{\sec x}_{\rightarrow +\infty} - \underbrace{\tan x}_{\rightarrow +\infty} \right) &= \lim_{x \rightarrow \frac{\pi}{2}^-} \underbrace{\frac{1 - \sin x}{\cos x}}_{\substack{\text{num} \rightarrow 0 \\ \text{den} \rightarrow 0}} = \lim_{x \rightarrow \frac{\pi}{2}^-} \underbrace{\frac{-\cos x}{-\sin x}}_{\substack{\text{num} \rightarrow 0 \\ \text{den} \rightarrow -1}} = 0
 \end{aligned}$$

Example 3.7.17

In the last example, Example 3.7.17 we converted an $\infty - \infty$ indeterminate form into a $\frac{0}{0}$ indeterminate form by exploiting the fact that the two terms, $\sec x$ and $\tan x$, in the $\infty - \infty$ indeterminate form shared a common denominator, namely $\cos x$. In the “real world” that will, of course, almost never happen. However as the next couple of examples show, you can often massage these expressions into suitable forms.

Here is another, much more complicated, example, where it doesn't happen.

Example 3.7.18

In this example, we evaluate the $\infty - \infty$ indeterminate form

$$\lim_{x \rightarrow 0} \left(\underbrace{\frac{1}{x}}_{\rightarrow \pm\infty} - \underbrace{\frac{1}{\log(1+x)}}_{\rightarrow \pm\infty} \right)$$

We convert it into a $\frac{0}{0}$ indeterminate form simply by putting the two fractions, $\frac{1}{x}$ and $\frac{1}{\log(1+x)}$ over a common denominator.

$$\lim_{x \rightarrow 0} \left(\underbrace{\frac{1}{x}}_{\rightarrow \pm\infty} - \underbrace{\frac{1}{\log(1+x)}}_{\rightarrow \pm\infty} \right) = \lim_{x \rightarrow 0} \underbrace{\frac{\log(1+x) - x}{x \log(1+x)}}_{\substack{\text{num} \rightarrow 0 \\ \text{den} \rightarrow 0}} \quad (\text{E1})$$

Now we apply L'Hôpital's rule, and simplify

$$\begin{aligned} \lim_{x \rightarrow 0} \underbrace{\frac{\log(1+x) - x}{x \log(1+x)}}_{\substack{\text{num} \rightarrow 0 \\ \text{den} \rightarrow 0}} &= \lim_{x \rightarrow 0} \frac{\frac{1}{1+x} - 1}{\log(1+x) + \frac{x}{1+x}} = \lim_{x \rightarrow 0} \frac{1 - (1+x)}{(1+x) \log(1+x) + x} \\ &= - \lim_{x \rightarrow 0} \underbrace{\frac{x}{(1+x) \log(1+x) + x}}_{\substack{\text{num} \rightarrow 0 \\ \text{den} \rightarrow 1 \times 0 + 0 = 0}} \quad (\text{E2}) \end{aligned}$$

Then we apply L'Hôpital's rule a second time

$$- \lim_{x \rightarrow 0} \underbrace{\frac{x}{(1+x) \log(1+x) + x}}_{\substack{\text{num} \rightarrow 0 \\ \text{den} \rightarrow 1 \times 0 + 0 = 0}} = - \lim_{x \rightarrow 0} \underbrace{\frac{1}{\log(1+x) + \frac{1+x}{1+x} + 1}}_{\substack{\text{num} \rightarrow 1 \\ \text{den} \rightarrow 0 + 1 + 1 = 2}} = -\frac{1}{2} \quad (\text{E3})$$

Combining (E1), (E2) and (E3) gives our final answer

$$\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{\log(1+x)} \right) = -\frac{1}{2}$$

Example 3.7.18

The following example can be done by l'Hôpital's rule, but it is actually far simpler to multiply by the conjugate and take the limit using the tools of Chapter 1.

Example 3.7.19

Consider the limit

$$\lim_{x \rightarrow \infty} \sqrt{x^2 + 4x} - \sqrt{x^2 - 3x}$$

Neither term is a fraction, but we can write

$$\begin{aligned} \sqrt{x^2 + 4x} - \sqrt{x^2 - 3x} &= x\sqrt{1 + 4/x} - x\sqrt{1 - 3/x} && \text{assuming } x > 0 \\ &= x \left(\sqrt{1 + 4/x} - \sqrt{1 - 3/x} \right) \\ &= \frac{\sqrt{1 + 4/x} - \sqrt{1 - 3/x}}{1/x} \end{aligned}$$

which is now a $0/0$ form with $f(x) = \sqrt{1 + 4/x} - \sqrt{1 - 3/x}$ and $g(x) = 1/x$. Then

$$f'(x) = \frac{-4/x^2}{2\sqrt{1 + 4/x}} - \frac{3/x^2}{2\sqrt{1 - 3/x}} \qquad g'(x) = -\frac{1}{x^2}$$

Hence

$$\frac{f'(x)}{g'(x)} = \frac{4}{2\sqrt{1 + 4/x}} + \frac{3}{\sqrt{1 - 3/x}}$$

And so in the limit as $x \rightarrow \infty$

$$\lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)} = \frac{4}{2} + \frac{3}{2} = \frac{7}{2}$$

and so our original limit is also $7/2$.

By comparison, if we multiply by the conjugate we have

$$\begin{aligned} \sqrt{x^2 + 4x} - \sqrt{x^2 - 3x} &= \left(\sqrt{x^2 + 4x} - \sqrt{x^2 - 3x} \right) \cdot \frac{\sqrt{x^2 + 4x} + \sqrt{x^2 - 3x}}{\sqrt{x^2 + 4x} + \sqrt{x^2 - 3x}} \\ &= \frac{x^2 + 4x - (x^2 - 3x)}{\sqrt{x^2 + 4x} + \sqrt{x^2 - 3x}} \\ &= \frac{7x}{\sqrt{x^2 + 4x} + \sqrt{x^2 - 3x}} \\ &= \frac{7}{\sqrt{1 + 4/x} + \sqrt{1 - 3/x}} && \text{assuming } x > 0 \end{aligned}$$

Now taking the limit as $x \rightarrow \infty$ gives $7/2$ as required. Just because we know l'Hôpital's rule, it does not mean we should use it everywhere it might be applied.

Example 3.7.19

(e) 1^∞ indeterminate form: We can use l'Hôpital's rule on limits of the form

$$\lim_{x \rightarrow a} f(x)^{g(x)} \text{ with } \lim_{x \rightarrow a} f(x) = 1 \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = \infty$$

by considering the logarithm of the limit⁶⁸:

$$\log \left(\lim_{x \rightarrow a} f(x)^{g(x)} \right) = \lim_{x \rightarrow a} \log \left(f(x)^{g(x)} \right) = \lim_{x \rightarrow a} \log (f(x)) \cdot g(x)$$

which is now an $0 \cdot \infty$ form. This can be further transformed into a $0/0$ or ∞/∞ form:

$$\begin{aligned} \log \left(\lim_{x \rightarrow a} f(x)^{g(x)} \right) &= \lim_{x \rightarrow a} \log (f(x)) \cdot g(x) \\ &= \lim_{x \rightarrow a} \frac{\log (f(x))}{1/g(x)}. \end{aligned}$$

Example 3.7.20

The following limit appears quite naturally when considering systems which display exponential growth or decay.

$$\lim_{x \rightarrow 0} (1+x)^{a/x} \quad \text{with the constant } a \neq 0$$

Since $(1+x) \rightarrow 1$ and $a/x \rightarrow \infty$ this is an 1^∞ indeterminate form.

By considering its logarithm we have

$$\begin{aligned} \log \left(\lim_{x \rightarrow 0} (1+x)^{a/x} \right) &= \lim_{x \rightarrow 0} \log \left((1+x)^{a/x} \right) \\ &= \lim_{x \rightarrow 0} \frac{a}{x} \log(1+x) \\ &= \lim_{x \rightarrow 0} \frac{a \log(1+x)}{x} \end{aligned}$$

which is now a $0/0$ form. Applying l'Hôpital's rule gives

$$\lim_{x \rightarrow 0} \underbrace{\frac{a \log(1+x)}{x}}_{\substack{\text{num} \rightarrow 0 \\ \text{den} \rightarrow 0}} = \lim_{x \rightarrow 0} \underbrace{\frac{\frac{a}{1+x}}{1}}_{\substack{\text{num} \rightarrow a \\ \text{den} \rightarrow 1}} = a$$

Since $(1+x)^{a/x} = \exp \left[\log \left((1+x)^{a/x} \right) \right]$ and the exponential function is continuous, our original limit is e^a .

Example 3.7.20

68 We are using the fact that the logarithm is a continuous function and Theorem 1.6.10

Here is a more complicated example of a 1^∞ indeterminate form.

Example 3.7.21

In the limit

$$\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{1/x^2}$$

the base, $\frac{\sin x}{x}$, converges to 1 (see Example 3.7.3) and the exponent, $\frac{1}{x^2}$, goes to ∞ . But if we take logarithms then

$$\log \left(\frac{\sin x}{x} \right)^{1/x^2} = \frac{\log \frac{\sin x}{x}}{x^2}$$

then, in the limit $x \rightarrow 0$, we have a $0/0$ indeterminate form. One application of l'Hôpital's rule gives

$$\lim_{x \rightarrow 0} \underbrace{\frac{\log \frac{\sin x}{x}}{x^2}}_{\substack{\text{num} \rightarrow 0 \\ \text{den} \rightarrow 0}} = \lim_{x \rightarrow 0} \frac{\frac{x}{\sin x} \frac{x \cos x - \sin x}{x^2}}{2x} = \lim_{x \rightarrow 0} \frac{\frac{x \cos x - \sin x}{x \sin x}}{2x} = \lim_{x \rightarrow 0} \frac{x \cos x - \sin x}{2x^2 \sin x}$$

which is another $0/0$ form. Applying l'Hôpital's rule again gives:

$$\begin{aligned} \lim_{x \rightarrow 0} \underbrace{\frac{x \cos x - \sin x}{2x^2 \sin x}}_{\substack{\text{num} \rightarrow 0 \\ \text{den} \rightarrow 0}} &= \lim_{x \rightarrow 0} \frac{\cos x - x \sin x - \cos x}{4x \sin x + 2x^2 \cos x} \\ &= - \lim_{x \rightarrow 0} \frac{x \sin x}{4x \sin x + 2x^2 \cos x} = - \lim_{x \rightarrow 0} \frac{\sin x}{4 \sin x + 2x \cos x} \end{aligned}$$

which is yet another $0/0$ form. Once more with l'Hôpital's rule:

$$\begin{aligned} - \lim_{x \rightarrow 0} \underbrace{\frac{\sin x}{4 \sin x + 2x \cos x}}_{\substack{\text{num} \rightarrow 0 \\ \text{den} \rightarrow 0}} &= - \lim_{x \rightarrow 0} \underbrace{\frac{\cos x}{4 \cos x + 2 \cos x - 2x \sin x}}_{\substack{\text{num} \rightarrow 1 \\ \text{den} \rightarrow 6}} \\ &= -\frac{1}{6} \end{aligned}$$

Oof! We have just shown that the logarithm of our original limit is $-1/6$. Hence the original limit itself is $e^{-1/6}$.

This was quite a complicated example. However it does illustrate the importance of cleaning up your algebraic expressions. This will both reduce the amount of work you have to do and will also reduce the number of errors you make.

Example 3.7.21

- (f) 0^0 indeterminate form: Like the 1^∞ form, this can be treated by considering its logarithm.

Example 3.7.22

For example, in the limit

$$\lim_{x \rightarrow 0^+} x^x$$

both the base, x , and the exponent, also x , go to zero. But if we consider the logarithm then we have

$$\log x^x = x \log x$$

which is a $0 \cdot \infty$ indeterminate form, which we already know how to treat. In fact, we already found, in Example 3.7.15, that

$$\lim_{x \rightarrow 0^+} x \log x = 0$$

Since the exponential is a continuous function

$$\lim_{x \rightarrow 0^+} x^x = \lim_{x \rightarrow 0^+} \exp(x \log x) = \exp\left(\lim_{x \rightarrow 0^+} x \log x\right) = e^0 = 1$$

Example 3.7.22

- (g) ∞^0 indeterminate form: Again, we can treat this form by considering its logarithm.

Example 3.7.23

For example, in the limit

$$\lim_{x \rightarrow +\infty} x^{1/x}$$

the base, x , goes to infinity and the exponent, $\frac{1}{x}$, goes to zero. But if we take logarithms

$$\log x^{1/x} = \frac{\log x}{x}$$

which is an ∞/∞ form, which we know how to treat.

$$\lim_{x \rightarrow +\infty} \underbrace{\frac{\log x}{x}}_{\substack{\text{num} \rightarrow \infty \\ \text{den} \rightarrow \infty}} = \lim_{x \rightarrow +\infty} \underbrace{\frac{\frac{1}{x}}{1}}_{\substack{\text{num} \rightarrow 0 \\ \text{den} \rightarrow 1}} = 0$$

Since the exponential is a continuous function

$$\lim_{x \rightarrow +\infty} x^{1/x} = \lim_{x \rightarrow +\infty} \exp\left(\frac{\log x}{x}\right) = \exp\left(\lim_{x \rightarrow +\infty} \frac{\log x}{x}\right) = e^0 = 1$$

Example 3.7.23

TOWARDS INTEGRAL CALCULUS

4.1 ▲ Introduction to Antiderivatives

We have now come to the final topic of the course — antiderivatives. This is only a short section since it is really just to give a taste of the next calculus subject: integral calculus.

So far in the course we have learned how to determine the rate of change (i.e. the derivative) of a given function. That is

given a function $f(x)$ find $\frac{df}{dx}$.

Along the way we developed an understanding of limits, which allowed us to define instantaneous rates of change — the derivative. We then went on to develop a number of applications of derivatives to modelling and approximation. In this last section we want to just introduce the idea of antiderivatives. That is

given a derivative $\frac{df}{dx}$ find the original function $f(x)$.

For example — say we know that

$$\frac{df}{dx} = x^2$$

and we want to find $f(x)$. From our previous experience differentiating we know that derivatives of polynomials are again polynomials. So we guess that our unknown function $f(x)$ is a polynomial. Further we know that when we differentiate x^n we get nx^{n-1} — multiply by the exponent and reduce the exponent by 1. So to end up with a derivative of x^2 we need to have differentiated an x^3 . But $\frac{d}{dx}x^3 = 3x^2$, so we need to divide both sides by 3 to get the answer we want. That is

$$\frac{d}{dx} \left(\frac{1}{3}x^3 \right) = x^2$$

However we know that the derivative of a constant is zero, so we also have

$$\frac{d}{dx} \left(\frac{1}{3}x^3 + 1 \right) = x^2$$

and

$$\frac{d}{dx} \left(\frac{1}{3}x^3 - \pi \right) = x^2$$

At this point it will really help the discussion to give a name to what we are doing.

Definition 4.1.1.

A function $F(x)$ that satisfies

$$\frac{d}{dx} F(x) = f(x)$$

is called an antiderivative of $f(x)$.

Notice the use of the indefinite article there — *an* antiderivative. This is precisely because we can always add or subtract a constant to an antiderivative and when we differentiate we'll get the same answer. We can write this as a lemma, but it is actually just Corollary [2.13.12](#) (from back in the section on the mean-value theorem) in disguise.

Lemma 4.1.2.

Let $F(x)$ be an antiderivative of $f(x)$, then for any constant c , the function $F(x) + c$ is also an antiderivative of $f(x)$.

Because of this lemma we typically write antiderivatives with “ $+c$ ” tacked on the end. That is, if we know that $F'(x) = f(x)$, then we would state that *the* antiderivative of $f(x)$ is

$$F(x) + c$$

where this “ $+c$ ” is there to remind us that we can always add or subtract some constant and it will still be an antiderivative of $f(x)$. Hence the antiderivative of x^2 is

$$\frac{1}{3}x^3 + c$$

Similarly, the antiderivative of x^4 is

$$\frac{1}{5}x^5 + c$$

and for $\sqrt{x} = x^{1/2}$ it is

$$\frac{2}{3}x^{3/2} + c$$

This last one is tricky (at first glance) — but we can always check our answer by differentiating.

$$\frac{d}{dx} \left(\frac{2}{3}x^{3/2} + c \right) = \frac{2}{3} \cdot \frac{3}{2}x^{1/2} + 0 \quad \checkmark$$

Now in order to determine the value of c we need more information. For example, we might be asked

Given that $g'(t) = t^2$ and $g(3) = 7$ find $g(t)$.

We are given the derivative and one piece of additional information and from these two facts we need to find the original function. From our work above we know that

$$g(t) = \frac{1}{3}t^3 + c$$

and we can find c from the other piece of information

$$7 = g(3) = \frac{1}{3} \cdot 27 + c = 9 + c$$

Hence $c = -2$ and so

$$g(t) = \frac{1}{3}t^3 - 2$$

We can then very easily check our answer by recomputing $g(3)$ and $g'(t)$. This is a good habit to get into.

Finding antiderivatives of polynomials is generally not too hard. We just need to use the rule

$$\text{if } f(x) = x^n \text{ then } F(x) = \frac{1}{n+1}x^{n+1} + c.$$

Of course this breaks down when $n = -1$. In order to find an antiderivative for $f(x) = \frac{1}{x}$ we need to remember that $\frac{d}{dx} \log x = \frac{1}{x}$, and more generally that $\frac{d}{dx} \log |x| = \frac{1}{x}$. See Example [2.10.4](#). So

$$\text{if } f(x) = \frac{1}{x} \text{ then } F(x) = \log |x| + c$$

Note that both $\frac{1}{x}$ and $\log |x|$ are not defined at $x = 0$. If you are interested, for example, in functions that are defined all $x \neq 0$, then, for any constants c_1 and c_2 ,

$$F(x) = \begin{cases} \log |x| + c_1 & \text{if } x > 0 \\ \log |x| + c_2 & \text{if } x < 0 \end{cases}$$

obeys $F'(x) = \frac{1}{x}$ for all $x \neq 0$. However, typically, in applications, either all of the x 's of interest are strictly positive, or all of the x 's of interest are negative. In both of those cases, the general $F(x)$ is $\log|x| + c$. So, typically, in tables of antiderivatives, the antiderivative of $1/x$ is listed as $\log|x| + c$.

Example 4.1.3

Let $f(x) = 3x^5 - 7x^2 + 2x + 3 + x^{-1} - x^{-2}$. Then the antiderivative of $f(x)$ is

$$\begin{aligned} F(x) &= \frac{3}{6}x^6 - \frac{7}{3}x^3 + \frac{2}{2}x^2 + 3x + \log|x| - \frac{1}{-1}x^{-1} + c && \text{clean it up} \\ &= \frac{1}{2}x^6 - \frac{7}{3}x^3 + x^2 + 3x + \log|x| + x^{-1} + c \end{aligned}$$

Now to check we should differentiate and hopefully we get back to where we started

$$\begin{aligned} F'(x) &= \frac{6}{2}x^5 - \frac{7}{3} \cdot 3x^2 + 2x + 3 + \frac{1}{x} - x^{-2} \\ &= 3x^5 - 7x^2 + 2x + 3 + \frac{1}{x} - x^{-2} \quad \checkmark \end{aligned}$$

Example 4.1.3

In your next calculus course you will develop a lot of machinery to help you find antiderivatives. At this stage about all that we can do is continue the sort of thing we have done. Think about the derivatives we know and work backwards. So, for example, we can take a list of derivatives

$F(x)$	1	x^n	$\sin x$	$\cos x$	$\tan x$	e^x	$\ln x $	$\arcsin x$	$\arctan x$
$f(x) = \frac{d}{dx}F(x)$	0	nx^{n-1}	$\cos x$	$-\sin x$	$\sec^2 x$	e^x	$\frac{1}{x}$	$\frac{1}{\sqrt{1-x^2}}$	$\frac{1}{1+x^2}$

and flip it upside down to give the tables

$f(x) = \frac{d}{dx}F(x)$	0	nx^{n-1}	$\cos x$	$-\sin x$	$\sec^2 x$	e^x	$\frac{1}{x}$
$F(x)$	c	$x^n + c$	$\sin x + c$	$\cos x + c$	$\tan x + c$	$e^x + c$	$\ln x + c$

$f(x) = \frac{d}{dx}F(x)$	$\frac{1}{\sqrt{1-x^2}}$	$\frac{1}{1+x^2}$
$F(x)$	$\arcsin x + c$	$\arctan x + c$

of antiderivatives. Here c is just a constant — any constant. But we can do a little more; clean up x^n by dividing by n and then replacing n by $n+1$. Similarly we can tweak $\sin x$ by multiplying by -1 :

$f(x) = \frac{d}{dx}F(x)$	0	x^n	$\cos x$	$\sin x$	$\sec^2 x$	e^x	$\frac{1}{x}$
$F(x)$	c	$\frac{1}{n+1}x^{n+1} + c$	$\sin x + c$	$-\cos x + c$	$\tan x + c$	$e^x + c$	$\ln x + c$

$f(x) = \frac{d}{dx}F(x)$	$\frac{1}{\sqrt{1-x^2}}$	$\frac{1}{1+x^2}$
$F(x)$	$\arcsin x + c$	$\arctan x + c$

Here are a couple more examples.

Example 4.1.4

Consider the functions

$$f(x) = \sin x + \cos 2x \qquad g(x) = \frac{1}{1+4x^2}$$

Find their antiderivatives.

Solution. The first one we can almost just look up our table. Let F be the antiderivative of f , then

$$F(x) = -\cos x + \sin 2x + c \qquad \text{is not quite right.}$$

When we differentiate to check things, we get a factor of two coming from the chain rule. Hence to compensate for that we multiply $\sin 2x$ by $\frac{1}{2}$:

$$F(x) = -\cos x + \frac{1}{2} \sin 2x + c$$

Differentiating this shows that we have the right answer.

Similarly, if we use G to denote the antiderivative of g , then it appears that G is nearly $\arctan x$. To get this extra factor of 4 we need to substitute $x \mapsto 2x$. So we try

$$G(x) = \arctan(2x) + c \qquad \text{which is nearly correct.}$$

Differentiating this gives us

$$G'(x) = \frac{2}{1+(2x)^2} = 2g(x)$$

Hence we should multiply by $\frac{1}{2}$. This gives us

$$G(x) = \frac{1}{2} \arctan(2x) + c.$$

We can then check that this is, in fact, correct just by differentiating.

Example 4.1.4

Now let's do a more substantial example.

Example 4.1.5

Suppose that we are driving to class. We start at $x = 0$ at time $t = 0$. Our velocity is $v(t) = 50 \sin(t)$. The class is at $x = 25$. When do we get there?

Solution. Let's denote by $x(t)$ our position at time t . We are told that

- $x(0) = 0$
- $x'(t) = 50 \sin t$

We have to determine $x(t)$ and then find the time T that obeys $x(T) = 25$. Now armed with our table above we know that the antiderivative of $\sin t$ is just $-\cos t$. We can check this:

$$\frac{d}{dt}(-\cos t) = \sin t$$

We can then get the factor of 50 by multiplying both sides of the above equation by 50:

$$\frac{d}{dt}(-50 \cos t) = 50 \sin t$$

And of course, this is just *an* antiderivative of $50 \sin t$; to write down the general antiderivative we just add a constant c :

$$\frac{d}{dt}(-50 \cos t + c) = 50 \sin t$$

Since $v(t) = \frac{d}{dt}x(t)$, this antiderivative is $x(t)$:

$$x(t) = -50 \cos t + c$$

To determine c we make use of the other piece of information we are given, namely

$$x(0) = 0.$$

Substituting this in gives us

$$x(0) = -50 \cos 0 + c = -50 + c$$

Hence we must have $c = 50$ and so

$$x(t) = -50 \cos t + 50 = 50(1 - \cos t).$$

Now that we have our position as a function of time, we can determine how long it takes us to arrive there. That is, we can find the time T so that $x(T) = 25$.

$$25 = x(T) = 50(1 - \cos T) \quad \text{so}$$

$$\frac{1}{2} = 1 - \cos T$$

$$-\frac{1}{2} = -\cos T$$

$$\frac{1}{2} = \cos T.$$

Recalling our special triangles, we see that $T = \frac{\pi}{3}$.

Example 4.1.5

The example below shows how antiderivatives arise naturally when studying differential equations.

Example 4.1.6 (Theorem 3.3.2 revisited.)

Back in Section 3.3 we encountered a simple differential equation, namely equation 3.3.1. We were able to solve this equation by guessing the answer and then checking it carefully. We can derive the solution more systematically by using antiderivatives.

Recall equation 3.3.1:

$$\frac{dQ}{dt} = -kQ$$

where $Q(t)$ is the amount of radioactive material at time t and we assume $Q(t) > 0$. Take this equation and divide both sides by $Q(t)$ to get

$$\frac{1}{Q(t)} \frac{dQ}{dt} = -k$$

At this point we should¹ think that the left-hand side is familiar. Now is a good moment to look back at logarithmic differentiation in Section 2.10.

The left-hand side is just the derivative of $\log Q(t)$:

$$\begin{aligned} \frac{d}{dt} (\log Q(t)) &= \frac{1}{Q(t)} \frac{dQ}{dt} \\ &= -k \end{aligned}$$

So to solve this equation, we are really being asked to find all functions $\log Q(t)$ having derivative $-k$. That is, we need to find all antiderivatives of $-k$. Of course that is just $-kt + c$. Hence we must have

$$\log Q(t) = -kt + c$$

and then taking the exponential of both sides gives

$$Q(t) = e^{-kt+c} = e^c \cdot e^{-kt} = Ce^{-kt}$$

where $C = e^c$. This is precisely Theorem 3.3.2.

Example 4.1.6

The above is a small example of the interplay between antiderivatives and differential equations.

Here is another example of how we might use antidifferentiation to compute areas or volumes.

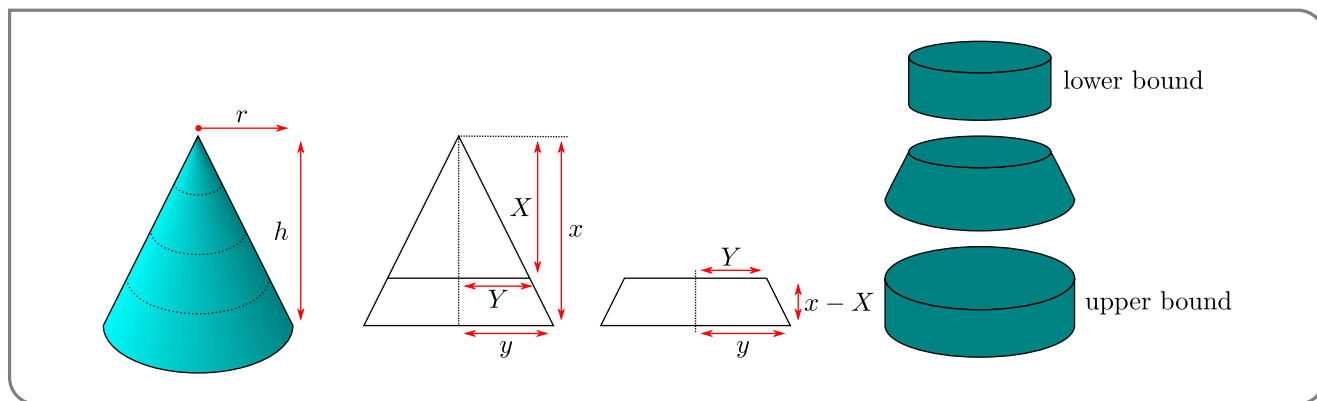
Example 4.1.7

We know (especially if one has revised the material in the appendix and Appendix B.5.2 in particular) that the volume of a right-circular cone is

$$V = \frac{\pi}{3} r^2 h$$

1 Well — perhaps it is better to say “notice that”. Let’s not make this a moral point.

where h is the height of the cone and r is the radius of its base. Now, the derivation of this formula given in Appendix [B.5.2](#) is not too simple. We present an alternate proof here that uses antiderivatives.



Consider cutting off a portion of the cone so that its new height is x (rather than h). Call the volume of the resulting smaller cone $V(x)$. We are going to determine $V(x)$ for all $x \geq 0$, including $x = h$, by first evaluating $V'(x)$ and $V(0)$ (which is obviously 0).

Call the radius of the base of the new smaller cone y (rather than r). By similar triangles we know that

$$\frac{r}{h} = \frac{y}{x}.$$

Now keep x and y fixed and consider cutting off a little more of the cone so its height is X . When we do so, the radius of the base changes from y to Y and again by similar triangles we know that

$$\frac{Y}{X} = \frac{y}{x} = \frac{r}{h}$$

The change in volume is then

$$V(x) - V(X)$$

Of course if we knew the formula for the volume of a cone, then we could compute the above exactly. However, even without knowing the volume of a cone, it is easy to derive upper and lower bounds on this quantity. The piece removed has bottom radius y and top radius Y . Hence its volume is bounded above and below by the cylinders of height $x - X$ and with radius y and Y respectively. Hence

$$\pi Y^2(x - X) \leq V(x) - V(X) \leq \pi y^2(x - X)$$

since the volume of a cylinder is just the area of its base times its height. Now massage this expression a little

$$\pi Y^2 \leq \frac{V(x) - V(X)}{x - X} \leq \pi y^2$$

The middle term now looks like a derivative; all we need to do is take the limit as $X \rightarrow x$:

$$\lim_{X \rightarrow x} \pi Y^2 \leq \lim_{X \rightarrow x} \frac{V(x) - V(X)}{x - X} \leq \lim_{X \rightarrow x} \pi y^2$$

The rightmost term is independent of X and so is just πy^2 . In the leftmost term, as $X \rightarrow x$, we must have that $Y \rightarrow y$. Hence the leftmost term is just πy^2 . Then by the squeeze theorem (Theorem [1.4.17](#)) we know that

$$\frac{dV}{dx} = \lim_{X \rightarrow x} \frac{V(x) - V(X)}{x - X} = \pi y^2.$$

But we know that

$$y = \frac{r}{h} \cdot x$$

so

$$\frac{dV}{dx} = \pi \left(\frac{r}{h} \right)^2 x^2$$

Now we can antidifferentiate to get back to V :

$$V(x) = \frac{\pi}{3} \left(\frac{r}{h} \right)^2 x^3 + c$$

To determine c notice that when $x = 0$ the volume of the cone is just zero and so $c = 0$. Thus

$$V(x) = \frac{\pi}{3} \left(\frac{r}{h} \right)^2 x^3$$

and so when $x = h$ we are left with

$$V(h) = \frac{\pi}{3} \left(\frac{r}{h} \right)^2 h^3 = \frac{\pi}{3} r^2 h$$

as required.

Example 4.1.7

You may use these properties in your solution to this question.

- (a) Sketch the graphs of $\sinh(x)$ and $\cosh(x)$.
- (b) Define inverse hyperbolic trigonometric functions $\sinh^{-1}(x)$ and $\cosh^{-1}(x)$, carefully specifying their domains of definition. Sketch the graphs of $\sinh^{-1}(x)$ and $\cosh^{-1}(x)$.
- (c) Find $\frac{d}{dx} \left\{ \cosh^{-1}(x) \right\}$.

3.7▲ L'Hôpital's Rule and indeterminate forms

Exercises

Jump to [HINTS](#), [ANSWERS](#), [SOLUTIONS](#) or [TABLE OF CONTENTS](#).

► Stage 1

In Questions [1](#) to [3](#) you are asked to give pairs of functions that combine to make indeterminate forms. Remember that an indeterminate form is indeterminate precisely because its limit can take on a number of values.

Q[1]: Give two functions $f(x)$ and $g(x)$ with the following properties:

- (i) $\lim_{x \rightarrow \infty} f(x) = \infty$
- (ii) $\lim_{x \rightarrow \infty} g(x) = \infty$
- (iii) $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = 2.5$

Q[2]: Give two functions $f(x)$ and $g(x)$ with the following properties:

- (i) $\lim_{x \rightarrow \infty} f(x) = \infty$
- (ii) $\lim_{x \rightarrow \infty} g(x) = \infty$
- (iii) $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = 0$

Q[3]: Give two functions $f(x)$ and $g(x)$ with the following properties:

- (i) $\lim_{x \rightarrow \infty} f(x) = 1$
- (ii) $\lim_{x \rightarrow \infty} g(x) = \infty$
- (iii) $\lim_{x \rightarrow \infty} [f(x)]^{g(x)} = 5$

► Stage 2

$$\text{Q[4](*)}: \text{Evaluate } \lim_{x \rightarrow 1} \frac{x^3 - e^{x-1}}{\sin(\pi x)}.$$

$$\text{Q[5](*)}: \text{Evaluate } \lim_{x \rightarrow 0^+} \frac{\log x}{x}. \text{ (Remember: in these notes, log means logarithm base } e \text{.)}$$

$$\text{Q[6](*)}: \text{Evaluate } \lim_{x \rightarrow \infty} (\log x)^2 e^{-x}.$$

$$\text{Q[7](*)}: \text{Evaluate } \lim_{x \rightarrow \infty} x^2 e^{-x}.$$

$$\text{Q[8](*)}: \text{Evaluate } \lim_{x \rightarrow 0} \frac{x - x \cos x}{x - \sin x}.$$

$$\text{Q[9]}: \text{Evaluate } \lim_{x \rightarrow 0} \frac{\sqrt{x^6 + 4x^4}}{x^2 \cos x}.$$

$$\text{Q[10](*)}: \text{Evaluate } \lim_{x \rightarrow \infty} \frac{(\log x)^2}{x}.$$

$$\text{Q[11](*)}: \text{Evaluate } \lim_{x \rightarrow 0} \frac{1 - \cos x}{\sin^2 x}.$$

$$\text{Q[12]}: \text{Evaluate } \lim_{x \rightarrow 0} \frac{x}{\sec x}.$$

$$\text{Q[13]}: \text{Evaluate } \lim_{x \rightarrow 0} \frac{\csc x \cdot \tan x \cdot (x^2 + 5)}{e^x}.$$

$$\text{Q[14]}: \text{Evaluate } \lim_{x \rightarrow \infty} \sqrt{2x^2 + 1} - \sqrt{x^2 + x}.$$

$$\text{Q[15](*)}: \text{Evaluate } \lim_{x \rightarrow 0} \frac{\sin(x^3 + 3x^2)}{\sin^2 x}.$$

$$\text{Q[16](*)}: \text{Evaluate } \lim_{x \rightarrow 1} \frac{\log(x^3)}{x^2 - 1}.$$

$$\text{Q[17](*)}: \text{Evaluate } \lim_{x \rightarrow 0} \frac{e^{-1/x^2}}{x^4}.$$

$$\text{Q[18](*)}: \text{Evaluate } \lim_{x \rightarrow 0} \frac{xe^x}{\tan(3x)}.$$

$$\text{Q[19]}: \text{Evaluate } \lim_{x \rightarrow 0} \sqrt[3]{\sin^2 x}.$$

Q[20]: Evaluate $\lim_{x \rightarrow 0} \sqrt[x]{\cos x}$.

Q[21]: Evaluate $\lim_{x \rightarrow 0^+} e^{x \log x}$.

Q[22]: Evaluate $\lim_{x \rightarrow 0} \left[-\log(x^2) \right]^x$.

Q[23](*) : Find c so that $\lim_{x \rightarrow 0} \frac{1 + cx - \cos x}{e^{x^2} - 1}$ exists.

Q[24](*) : Evaluate $\lim_{x \rightarrow 0} \frac{e^{k \sin(x^2)} - (1 + 2x^2)}{x^4}$, where k is a constant.

►► Stage 3

Q[25]: Suppose an algorithm, given an input with n variables, will terminate in at most $S(n) = 5n^4 - 13n^3 - 4n + \log(n)$ steps. A researcher writes that the algorithm will terminate in *roughly* at most $A(n) = 5n^4$ steps. Show that the percentage error involved in using $A(n)$ instead of $S(n)$ tends to zero as n gets very large. What happens to the absolute error?

Remark: this is a very common kind of approximation. When people deal with functions that give very large numbers, often they don't care about the *exact* large number—they only want a ballpark. So, a complicated function might be replaced by an easier function that doesn't give a large relative error. If you would like to know more about the ways people describe functions that give very large numbers, you can read about “Big O Notation” in Section 3.6.3 of the CLP101 notes.

TOWARDS INTEGRAL CALCULUS

4.1▲ Introduction to antiderivatives

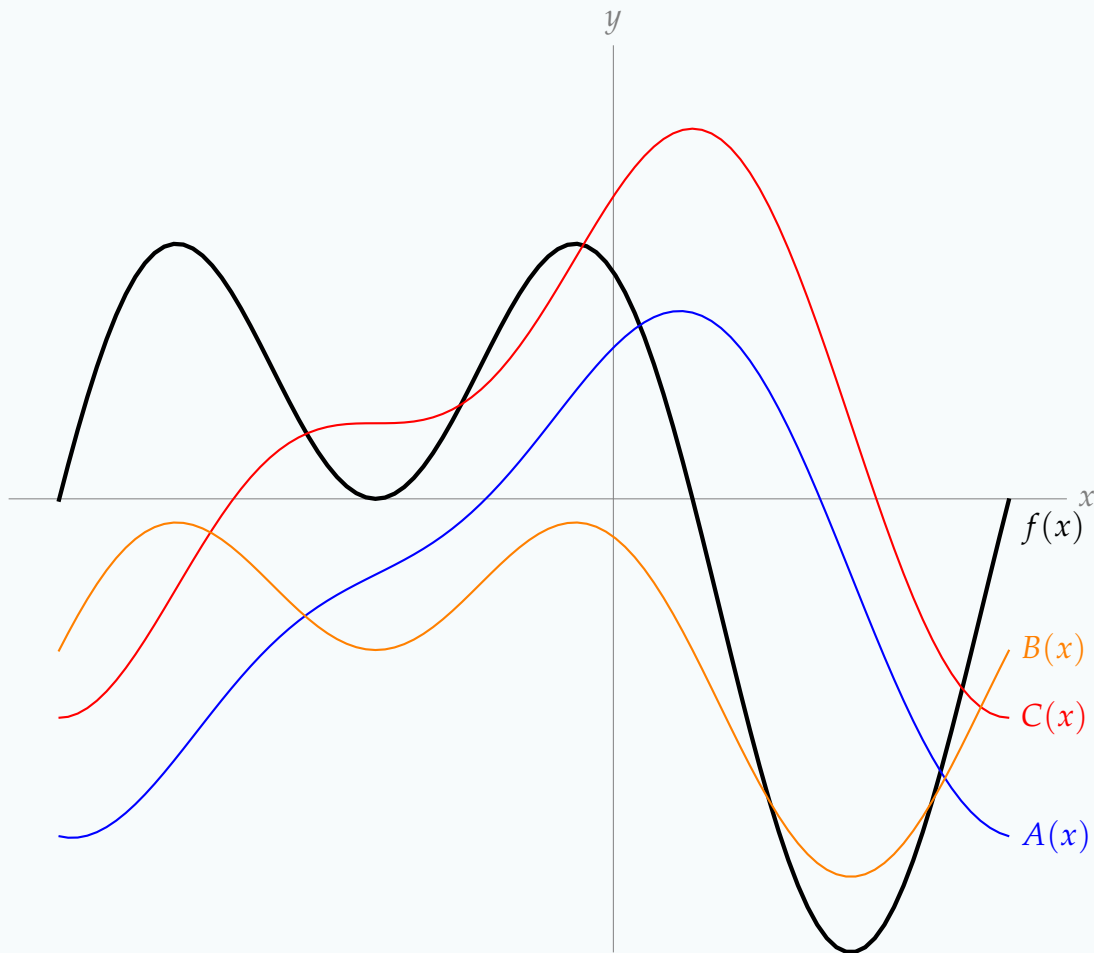
Exercises

Jump to [HINTS](#), [ANSWERS](#), [SOLUTIONS](#) or [TABLE OF CONTENTS](#).

►► Stage 1

Q[1]: Let $f(x)$ be a function with derivative $f'(x)$. What is the most general antiderivative of $f'(x)$?

Q[2]: On the graph below, the black curve is $y = f(x)$. Which of the coloured curves is an antiderivative of $f(x)$?



► Stage 2

In Questions [3](#) through [12](#), you are asked to find the antiderivative of a function. Phrased like this, we mean the most general antiderivative. These will all include some added constant. The table after Example 4.1.3 might be of help.

Q[3]: Find the antiderivative of $f(x) = 3x^2 + 5x^4 + 10x - 9$.

Q[4]: Find the antiderivative of $f(x) = \frac{3}{5}x^7 - 18x^4 + x$.

Q[5]: Find the antiderivative of $f(x) = 4\sqrt[3]{x} - \frac{9}{2x^{2.7}}$.

Q[6]: Find the antiderivative of $f(x) = \frac{1}{7\sqrt{x}}$.

Q[7]: Find the antiderivative of $f(x) = e^{5x+11}$.

Q[8]: Find the antiderivative of $f(x) = 3 \sin(5x) + 7 \cos(13x)$.

Q[9]: Find the antiderivative of $f(x) = \sec^2(x + 1)$.

Q[10]: Find the antiderivative of $f(x) = \frac{1}{x + 2}$.

Q[11]: Find the antiderivative of $f(x) = \frac{7}{\sqrt{3 - 3x^2}}$.

Q[12]: Find the antiderivative of $f(x) = \frac{1}{1 + 25x^2}$.

In Questions 13 through 16 you are asked to find a specific antiderivative of a function. In this case, you should be able to solve for the entire function—no unknown constants floating about.

Q[13]: Find the function $f(x)$ with $f'(x) = 3x^2 - 9x + 4$ and $f(1) = 10$.

Q[14]: Find the function $f(x)$ with $f'(x) = \cos(2x)$ and $f(\pi) = \pi$.

Q[15]: Find the function $f(x)$ that is defined and obeys $f'(x) = \frac{1}{x}$ for all $x < 0$, and also obeys $f(-1) = 0$.

Q[16]: Find the function $f(x)$ with $f'(x) = \frac{1}{\sqrt{1 - x^2}} + 1$ and $f(1) = -\frac{\pi}{2}$.

In Questions 17 through 19 we will explore some simple situations where antiderivatives might arise.

Q[17]: Suppose a population of bacteria at time t (measured in hours) is growing at a rate of $100e^{2t}$ individuals per hour. Starting at time $t = 0$, how long will it take the initial colony to increase by 300 individuals?

Q[18]: Your bank account at time t (measured in years) is growing at a rate of

$$1500e^{\frac{t}{50}}$$

dollars per year. How much money is in your account at time t ?

Q[19]: At time t during a particular day, $0 \leq t \leq 24$, your house consumes energy at a rate of

$$0.5 \sin\left(\frac{\pi}{24}t\right) + 0.25$$

kW. (Your consumption was smallest in the middle of the night, and peaked at noon.) How much energy did your house consume in that day?

► Stage 3

Q[20](*) : Let $f(x) = 2 \sin^{-1} \sqrt{x}$ and $g(x) = \sin^{-1}(2x - 1)$. Find the derivative of $f(x) - g(x)$ and simplify your answer. What does the answer imply about the relation between $f(x)$ and $g(x)$?

For Questions 21 through 26, you are again asked to find the antiderivatives of certain functions. In general, finding antiderivatives can be extremely difficult—indeed, it will form the main topic of next semester's calculus course. However, you can work out the antiderivatives of the functions below using what you've learned so far about derivatives.

Q[21]: Find the antiderivative of $f(x) = 2 \cos(2x) \cos(3x) - 3 \sin(2x) \sin(3x)$.

Q[22]: Find the antiderivative of $f(x) = \frac{(x^2 + 1)e^x - e^x(2x)}{(x^2 + 1)^2}$.

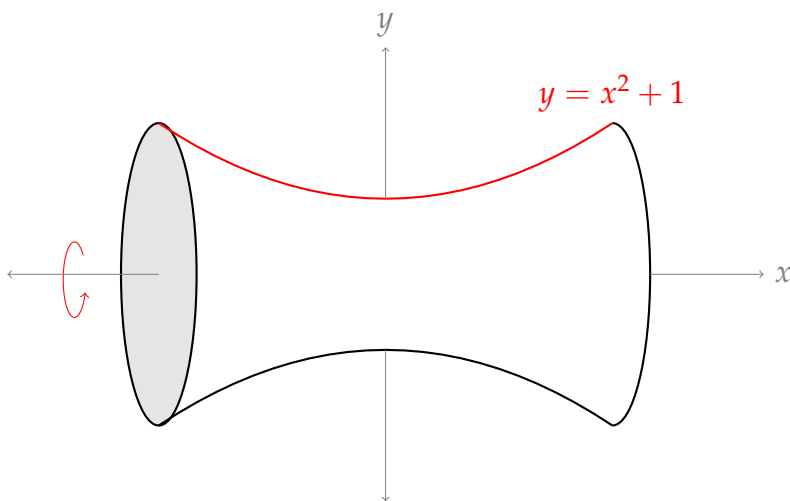
Q[23]: Find the antiderivative of $f(x) = 3x^2 e^{x^3}$.

Q[24]: Find the antiderivative of $f(x) = 5x \sin(x^2)$.

Q[25]: Find the antiderivative of $f(x) = e^{\log x}$.

Q[26]: Find the antiderivative of $f(x) = \frac{7}{\sqrt{3 - x^2}}$.

Q[27]: Imagine forming a solid by revolving the parabola $y = x^2 + 1$ around the x -axis, as in the picture below.



Use the method of Example 4.1.7 to find the volume of such an object if the segment of the parabola that we rotate runs from $x = -H$ to $x = H$.

Q[28]: Find all functions $f(x)$ that are defined and obey $f'(x) = \frac{1}{x}$ for all $x \neq 0$, and also obey $f(-1) = 0$.

Solutions to Exercises 3.7 — Jump to [TABLE OF CONTENTS](#)

S-1: There are many possible answers. Consider these: $f(x) = 5x$, $g(x) = 2x$. Then

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} g(x) = \infty, \text{ and } \lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} \frac{5x}{2x} = \lim_{x \rightarrow \infty} \frac{5}{2} = \frac{5}{2} = 2.5.$$

S-2: There are many possible answers. Consider these: $f(x) = x$, $g(x) = x^2$. Then

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} g(x) = \infty, \text{ and } \lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} \frac{x}{x^2} = \lim_{x \rightarrow \infty} \frac{1}{x} = 0.$$

S-3: From Example 3.7.20, we know that $\lim_{x \rightarrow 0} (1+x)^{\frac{a}{x}} = e^a$, so $\lim_{x \rightarrow 0} (1+x)^{\frac{\log 5}{x}} = e^{\log 5} = 5$. However, this is the limit as x goes to 0, which is not what we were asked. So, we modify the functions by replacing x with $\frac{1}{x}$. If $x \rightarrow 0^+$, then $\frac{1}{x} \rightarrow \infty$.

Taking $f(x) = 1 + \frac{1}{x}$ and $g(x) = x \log 5$, we see:

(i) $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right) = 1$

(ii) $\lim_{x \rightarrow \infty} g(x) = \lim_{x \rightarrow \infty} x \log 5 = \infty$

(iii) Let us name $\frac{1}{x} = X$. Then as $x \rightarrow \infty$, $X \rightarrow 0^+$, so:

$$\lim_{x \rightarrow \infty} [f(x)]^{g(x)} = \lim_{x \rightarrow \infty} \left[1 + \frac{1}{x}\right]^{x \log 5} = \lim_{x \rightarrow \infty} \left[1 + \frac{1}{x}\right]^{\frac{\log 5}{\frac{1}{x}}} = \lim_{X \rightarrow 0^+} [1 + X]^{\frac{\log 5}{X}} = e^{\log 5} = 5, \text{ where in the penultimate step, we used the result of Example 3.7.20.}$$

S-4: If we plug in $x = 1$ to the numerator and the denominator, we find they are both zero. So, we have an indeterminate form appropriate for L'Hôpital's Rule.

$$\lim_{x \rightarrow 1} \underbrace{\frac{x^3 - e^{x-1}}{\sin(\pi x)}}_{\substack{\text{num} \rightarrow 0 \\ \text{den} \rightarrow 0}} = \lim_{x \rightarrow 1} \frac{3x^2 - e^{x-1}}{\pi \cos(\pi x)} = -\frac{2}{\pi}$$

S-5: Be careful— this is not an indeterminate form!

As $x \rightarrow 0^+$, the numerator $\log x \rightarrow -\infty$. That is, the numerator is becoming an increasingly huge, negative number. As $x \rightarrow 0^+$, the denominator $x \rightarrow 0^+$, which only serves to make the total fraction even larger, and still negative. So, $\lim_{x \rightarrow 0^+} \frac{\log x}{x} = -\infty$.

Remark: if we had tried to use l'Hôpital's Rule here, we would have come up with the wrong answer. If we differentiate the numerator and the denominator, the fraction becomes $\frac{\frac{1}{x}}{1} = \frac{1}{x}$, and $\lim_{x \rightarrow 0^+} \frac{1}{x} = \infty$. The reason we cannot apply l'Hôpital's Rule is that we do not have an indeterminate form, like both numerator and denominator going to infinity, or both numerator and denominator going to zero.

S-6: We rearrange the expression to a more natural form:

$$\lim_{x \rightarrow \infty} (\log x)^2 e^{-x} = \lim_{x \rightarrow \infty} \underbrace{\frac{(\log x)^2}{e^x}}_{\substack{\text{num} \rightarrow \infty \\ \text{den} \rightarrow \infty}}$$

Both the numerator and denominator go to infinity as x goes to infinity. So, we can apply l'Hôpital's Rule. In fact, we end up applying it twice.

$$\begin{aligned} &= \lim_{x \rightarrow \infty} \underbrace{\frac{2 \log x}{x e^x}}_{\substack{\text{num} \rightarrow \infty \\ \text{den} \rightarrow \infty}} \\ &= \lim_{x \rightarrow \infty} \frac{2/x}{x e^x + e^x} \end{aligned}$$

The numerator gets smaller and smaller while the denominator gets larger and larger, so:

$$= 0$$

S-7:

$$\lim_{x \rightarrow \infty} x^2 e^{-x} = \lim_{x \rightarrow \infty} \underbrace{\frac{x^2}{e^x}}_{\substack{\text{num} \rightarrow \infty \\ \text{den} \rightarrow \infty}} = \lim_{x \rightarrow \infty} \underbrace{\frac{2x}{e^x}}_{\substack{\text{num} \rightarrow \infty \\ \text{den} \rightarrow \infty}} = \lim_{x \rightarrow \infty} \underbrace{\frac{2}{e^x}}_{\substack{\text{num} \rightarrow \infty \\ \text{den} \rightarrow \infty}} = 0$$

S-8:

$$\begin{aligned} \lim_{x \rightarrow 0} \underbrace{\frac{x - x \cos x}{x - \sin x}}_{\substack{\text{num} \rightarrow 0 \\ \text{den} \rightarrow 0}} &= \lim_{x \rightarrow 0} \underbrace{\frac{1 - \cos x + x \sin x}{1 - \cos x}}_{\substack{\text{num} \rightarrow 0 \\ \text{den} \rightarrow 0}} = \lim_{x \rightarrow 0} \underbrace{\frac{\sin x + \sin x + x \cos x}{\sin x}}_{\substack{\text{num} \rightarrow 0 \\ \text{den} \rightarrow 0}} \\ &= \lim_{x \rightarrow 0} \frac{2 \cos x + \cos x - x \sin x}{\cos x} = 3 \end{aligned}$$

S-9: If we plug in $x = 0$ to the numerator and denominator, both are zero, so this is a candidate for l'Hôpital's Rule. However, an easier way to evaluate the limit is to factor x^2 from the numerator and denominator, and cancel.

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sqrt{x^6 + 4x^4}}{x^2 \cos x} &= \lim_{x \rightarrow 0} \frac{\sqrt{x^4} \sqrt{x^2 + 4}}{x^2 \cos x} \\ &= \lim_{x \rightarrow 0} \frac{x^2 \sqrt{x^2 + 4}}{x^2 \cos x} \\ &= \lim_{x \rightarrow 0} \frac{\sqrt{x^2 + 4}}{\cos x} \\ &= \frac{\sqrt{0^2 + 4}}{\cos(0)} = 2 \end{aligned}$$

S-10:

$$\lim_{x \rightarrow \infty} \underbrace{\frac{(\log x)^2}{x}}_{\substack{\text{num} \rightarrow \infty \\ \text{den} \rightarrow \infty}} = \lim_{x \rightarrow \infty} \frac{2(\log x) \frac{1}{x}}{1} = 2 \lim_{x \rightarrow \infty} \underbrace{\frac{\log x}{x}}_{\substack{\text{num} \rightarrow \infty \\ \text{den} \rightarrow \infty}} = 2 \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{1} = 0$$

S-11:

$$\lim_{x \rightarrow 0} \underbrace{\frac{1 - \cos x}{\sin^2 x}}_{\substack{\text{num} \rightarrow 0 \\ \text{den} \rightarrow 0}} = \lim_{x \rightarrow 0} \frac{\sin x}{2 \sin x \cos x} = \lim_{x \rightarrow 0} \frac{1}{2 \cos x} = \frac{1}{2}$$

S-12: If we plug in $x = 0$, the numerator is zero, and the denominator is

$$\sec 0 = \frac{1}{\cos 0} = \frac{1}{1} = 1. \text{ So the limit is } \frac{0}{1} = 0.$$

Be careful: you cannot use l'Hôpital's Rule here, because the fraction does not give an indeterminate form. If you try to differentiate the numerator and the denominator, you get an expression whose limit does not exist:

$$\lim_{x \rightarrow 0} \frac{1}{\sec x \tan x} = \lim_{x \rightarrow 0} \cos x \cdot \frac{\cos x}{\sin x} = DNE.$$

S-13: If we plug $x = 0$ into the denominator, we get 1. However, the numerator is an indeterminate form: $\tan 0 = 0$, while $\lim_{x \rightarrow 0^+} \csc x = \infty$ and $\lim_{x \rightarrow 0^-} \csc x = -\infty$. If we use $\csc x = \frac{1}{\sin x}$, our expression becomes

$$\lim_{x \rightarrow 0} \frac{\tan x \cdot (x^2 + 5)}{\sin x \cdot e^x}$$

Since plugging in $x = 0$ makes both the numerator and the denominator equal to zero, this is a candidate for l'Hôpital's Rule. However, a much easier way is to simplify the trig first.

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\tan x \cdot (x^2 + 5)}{\sin x \cdot e^x} &= \lim_{x \rightarrow 0} \frac{\sin x \cdot (x^2 + 5)}{\cos x \cdot \sin x \cdot e^x} \\ &= \lim_{x \rightarrow 0} \frac{x^2 + 5}{\cos x \cdot e^x} \\ &= \frac{0^2 + 5}{\cos(0) \cdot e^0} = 5 \end{aligned}$$

S-14: $\lim_{x \rightarrow \infty} \sqrt{2x^2 + 1} - \sqrt{x^2 + x}$ has the indeterminate form $\infty - \infty$. To get a better idea of what's going on, let's rationalize.

$$\begin{aligned} \lim_{x \rightarrow \infty} \sqrt{2x^2 + 1} - \sqrt{x^2 + x} &= \lim_{x \rightarrow \infty} \left(\sqrt{2x^2 + 1} - \sqrt{x^2 + x} \right) \left(\frac{\sqrt{2x^2 + 1} + \sqrt{x^2 + x}}{\sqrt{2x^2 + 1} + \sqrt{x^2 + x}} \right) \\ &= \lim_{x \rightarrow \infty} \frac{(2x^2 + 1) - (x^2 + x)}{\sqrt{2x^2 + 1} + \sqrt{x^2 + x}} = \lim_{x \rightarrow \infty} \frac{x^2 - x + 1}{\sqrt{2x^2 + 1} + \sqrt{x^2 + x}} \end{aligned}$$

Here, we have the indeterminate form $\frac{\infty}{\infty}$, so l'Hôpital's Rule applies. However, if we try to use it here, we quickly get a huge mess. Instead, remember how we dealt with these kinds of limits in the past: factor out the highest power of the denominator, which is x .

$$\begin{aligned}
 &= \lim_{x \rightarrow \infty} \frac{x \left(x - 1 + \frac{1}{x} \right)}{\sqrt{x^2 \left(2 + \frac{1}{x^2} \right)} + \sqrt{x^2 \left(1 + \frac{1}{x} \right)}} \\
 &= \lim_{x \rightarrow \infty} \frac{x \left(x - 1 + \frac{1}{x} \right)}{x \left(\sqrt{2 + \frac{1}{x^2}} + \sqrt{1 + \frac{1}{x}} \right)} \\
 &= \lim_{x \rightarrow \infty} \frac{x - 1 + \frac{1}{x}}{\underbrace{\sqrt{2 + \frac{1}{x^2}} + \sqrt{1 + \frac{1}{x}}}_{\substack{\text{num} \rightarrow \infty \\ \text{den} \rightarrow \sqrt{2}+1}}} \\
 &= \infty
 \end{aligned}$$

S-15: If we plug in $x = 0$, both numerator and denominator become zero. So, we have exactly one of the indeterminate forms that l'Hôpital's Rule applies to.

$$\lim_{x \rightarrow 0} \underbrace{\frac{\sin(x^3 + 3x^2)}{\sin^2 x}}_{\substack{\text{num} \rightarrow 0 \\ \text{den} \rightarrow 0}} = \lim_{x \rightarrow 0} \frac{(3x^2 + 6x) \cos(x^3 + 3x^2)}{2 \sin x \cos x}$$

If we plug in $x = 0$, still we find that both the numerator and the denominator go to zero. We could jump in with another iteration of l'Hôpital's Rule. However, the derivatives would be a little messy, so we use limit laws and break up the fraction into the product of two fractions. If both limits exist:

$$\lim_{x \rightarrow 0} \frac{(3x^2 + 6x) \cos(x^3 + 3x^2)}{2 \sin x \cos x} = \left(\lim_{x \rightarrow 0} \frac{x^2 + 2x}{\sin x} \right) \cdot \left(\lim_{x \rightarrow 0} \frac{3 \cos(x^3 + 3x^2)}{2 \cos x} \right)$$

We can evaluate the right-hand limit by simply plugging in $x = 0$:

$$\begin{aligned}
 &= \frac{3}{2} \lim_{x \rightarrow 0} \underbrace{\frac{x^2 + 2x}{\sin x}}_{\substack{\text{num} \rightarrow 0 \\ \text{den} \rightarrow 0}} \\
 &= \frac{3}{2} \lim_{x \rightarrow 0} \frac{2x + 2}{\cos x} \\
 &= \frac{3}{2} \left(\frac{2}{1} \right) = 3
 \end{aligned}$$

S-16:

$$\lim_{x \rightarrow 1} \frac{\log(x^3)}{x^2 - 1} = \lim_{x \rightarrow 1} \underbrace{\frac{3 \log(x)}{x^2 - 1}}_{\substack{\text{num} \rightarrow 0 \\ \text{den} \rightarrow 0}} = \lim_{x \rightarrow 1} \frac{3/x}{2x} = \frac{3}{2}$$

S-17:

- Solution 1.

$$\lim_{x \rightarrow 0} \frac{e^{-1/x^2}}{x^4} = \lim_{x \rightarrow 0} \underbrace{\frac{\frac{1}{x^4}}{e^{1/x^2}}}_{\substack{\text{num} \rightarrow \infty \\ \text{den} \rightarrow \infty}} = \lim_{x \rightarrow 0} \frac{\frac{-4}{x^5}}{\frac{-2}{x^3} e^{1/x^2}} = \lim_{x \rightarrow 0} \underbrace{\frac{\frac{2}{x^2}}{e^{1/x^2}}}_{\substack{\text{num} \rightarrow \infty \\ \text{den} \rightarrow \infty}} = \lim_{x \rightarrow 0} \frac{\frac{-4}{x^3}}{\frac{-2}{x^3} e^{1/x^2}} = \lim_{x \rightarrow 0} \frac{2}{e^{1/x^2}} = 0$$

since, as $x \rightarrow 0$, the exponent $\frac{1}{x^2} \rightarrow \infty$ so that $e^{1/x^2} \rightarrow \infty$ and $e^{-1/x^2} \rightarrow 0$.

- Solution 2.

$$\lim_{x \rightarrow 0} \frac{e^{-1/x^2}}{x^4} = \lim_{t = \frac{1}{x^2} \rightarrow \infty} \frac{e^{-t}}{t^{-2}} = \lim_{t \rightarrow \infty} \underbrace{\frac{t^2}{e^t}}_{\substack{\text{num} \rightarrow \infty \\ \text{den} \rightarrow \infty}} = \lim_{t \rightarrow \infty} \underbrace{\frac{2t}{e^t}}_{\substack{\text{num} \rightarrow \infty \\ \text{den} \rightarrow \infty}} = \lim_{t \rightarrow \infty} \frac{2}{e^t} = 0$$

S-18:

$$\lim_{x \rightarrow 0} \underbrace{\frac{xe^x}{\tan(3x)}}_{\substack{\text{num} \rightarrow 0 \\ \text{den} \rightarrow 0}} = \lim_{x \rightarrow 0} \frac{e^x + xe^x}{3 \sec^2(3x)} = \frac{1}{3}$$

S-19: $\lim_{x \rightarrow 0} \sin^2 x = 0$, and $\lim_{x \rightarrow 0} \frac{1}{x^2} = \infty$, so we have the form 0^∞ . (Note that $\sin^2 x$ is positive, so our root is defined.) This is not an indeterminate form: $\lim_{x \rightarrow 0} \sqrt[2]{\sin^2 x} = 0$.

S-20: $\lim_{x \rightarrow 0} \cos x = 1$ and $\lim_{x \rightarrow 0} \frac{1}{x^2} = \infty$, so $\lim_{x \rightarrow 0} (\cos x)^{\frac{1}{x^2}}$ has the indeterminate form 1^∞ . We want to use l'Hôpital, but we need to get our function into a fractional indeterminate form. So, we'll use a logarithm.

$$\begin{aligned} y &:= (\cos x)^{\frac{1}{x^2}} \\ \log y &= \log \left((\cos x)^{\frac{1}{x^2}} \right) = \frac{1}{x^2} \log(\cos x) = \frac{\log \cos x}{x^2} \\ \lim_{x \rightarrow 0} \log y &= \lim_{x \rightarrow 0} \underbrace{\frac{\log \cos x}{x^2}}_{\substack{\text{num} \rightarrow 0 \\ \text{den} \rightarrow 0}} = \lim_{x \rightarrow 0} \frac{\frac{-\sin x}{\cos x}}{2x} = \lim_{x \rightarrow 0} \underbrace{\frac{-\tan x}{2x}}_{\substack{\text{num} \rightarrow 0 \\ \text{den} \rightarrow 0}} = \lim_{x \rightarrow 0} \frac{-\sec^2 x}{2} \\ &= \lim_{x \rightarrow 0} \frac{-1}{2 \cos^2 x} = -\frac{1}{2} \\ \text{Therefore, } \lim_{x \rightarrow 0} y &= \lim_{x \rightarrow 0} e^{\log y} = e^{-1/2} = \frac{1}{\sqrt{e}} \end{aligned}$$

S-21:

- Solution 1

$$\begin{aligned} y &:= e^{x \log x} = (e^x)^{\log x} \\ \lim_{x \rightarrow 0^+} y &= \lim_{x \rightarrow 0^+} (e^x)^{\log x} \end{aligned}$$

This has the form $1^{-\infty} = \frac{1}{1^\infty}$, and 1^∞ is an indeterminate form. We want to use l'Hôpital, but we need to get a different indeterminate form. So, we'll use logarithms.

$$\lim_{x \rightarrow 0^+} \log y = \lim_{x \rightarrow 0^+} \log \left((e^x)^{\log x} \right) = \lim_{x \rightarrow 0^+} \log x \log (e^x) = \lim_{x \rightarrow 0^+} (\log x) \cdot x$$

This has the indeterminate form $0 \cdot \infty$, so we need one last adjustment before we can use l'Hôpital's Rule.

$$= \lim_{x \rightarrow 0^+} \underbrace{\frac{\log x}{\frac{1}{x}}}_{\substack{\text{num} \rightarrow -\infty \\ \text{den} \rightarrow \infty}} = \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{\frac{-1}{x^2}} = \lim_{x \rightarrow 0^+} -x = 0$$

Now, we can figure out what happens to our original function, y :

$$\lim_{x \rightarrow 0^+} y = \lim_{x \rightarrow 0^+} e^{\log y} = e^0 = 1$$

• Solution 2

$$y := e^{x \log x} = \left(e^{\log x} \right)^x = x^x$$

$$\lim_{x \rightarrow 0^+} y = \lim_{x \rightarrow 0^+} x^x$$

We have the indeterminate form 0^0 . We want to use l'Hôpital, but we need a different indeterminate form. So, we'll use logarithms.

$$\lim_{x \rightarrow 0^+} \log y = \lim_{x \rightarrow 0^+} \log(x^x) = \lim_{x \rightarrow 0^+} x \log x$$

Now we have the indeterminate form $0 \cdot \infty$, so we need one last adjustment before we can use l'Hôpital's Rule.

$$\lim_{x \rightarrow 0^+} y = \lim_{x \rightarrow 0^+} \underbrace{\frac{\log x}{\frac{1}{x}}}_{\substack{\text{num} \rightarrow 0 \\ \text{den} \rightarrow -\infty}} = \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{\frac{-1}{x^2}} = \lim_{x \rightarrow 0^+} -x = 0$$

Now, we can figure out what happens to our original function, y :

$$\lim_{x \rightarrow 0^+} y = \lim_{x \rightarrow 0^+} e^{\log y} = e^0 = 1$$

S-22: First, note that the function exists near 0: x^2 is positive, so $\log(x^2)$ exists; near 0, $\log x^2$ is negative, so $-\log(x^2)$ is positive, so $[-\log(x^2)]^x$ exists even when x is negative.

Since $\lim_{x \rightarrow 0} -\log(x^2) = \infty$ and $\lim_{x \rightarrow 0} x = 0$, we have the indeterminate form ∞^0 . We need l'Hôpital, but we need to manipulate our function into an appropriate form. We do this using logarithms.

$$y := [-\log(x^2)]^x$$

$$\log y = \log \left([-\log(x^2)]^x \right) = \underbrace{x}_{\rightarrow 0} \cdot \underbrace{\log \left(\underbrace{-\log(x^2)}_{\rightarrow \infty} \right)}_{\rightarrow \infty} = \frac{\log(-\log(x^2))}{\frac{1}{x}}$$

$$\lim_{x \rightarrow 0} \log y = \lim_{x \rightarrow 0} \underbrace{\frac{\log(-\log(x^2))}{\frac{1}{x}}}_{\substack{\text{num} \rightarrow \infty \\ \text{den} \rightarrow \pm \infty}} = \lim_{x \rightarrow 0} \frac{\frac{-\frac{2}{x}}{-\log(x^2)}}{\frac{-1}{x^2}} = \lim_{x \rightarrow 0} \underbrace{\frac{-2x}{\log(x^2)}}_{\substack{\text{num} \rightarrow 0 \\ \text{den} \rightarrow -\infty}} = 0$$

Now, we're ready to figure out our original limit.

$$\lim_{x \rightarrow 0} y = \lim_{x \rightarrow 0} e^{\log y} = e^0 = 1$$

S-23: Both the numerator and denominator converge to 0 as $x \rightarrow 0$. So, by l'Hôpital,

$$\lim_{x \rightarrow 0} \underbrace{\frac{1 + cx - \cos x}{e^{x^2} - 1}}_{\substack{\text{num} \rightarrow 0 \\ \text{den} \rightarrow 0}} = \lim_{x \rightarrow 0} \frac{c + \sin x}{2xe^{x^2}}$$

The new denominator still converges to 0 as $x \rightarrow 0$. For the limit to exist, the same must be true for the new numerator. This tells us that if $c \neq 0$, the limit does not exist. We should check whether the limit exists when $c = 0$. Using l'Hôpital:

$$\lim_{x \rightarrow 0} \underbrace{\frac{\sin x}{2xe^{x^2}}}_{\substack{\text{num} \rightarrow 0 \\ \text{den} \rightarrow 0}} = \lim_{x \rightarrow 0} \frac{\cos x}{e^{x^2}(4x^2 + 2)} = \frac{1}{1(0 + 2)} = \frac{1}{2}.$$

So, the limit exists when $c = 0$.

S-24: The first thing we notice is, regardless of k , when we plug in $x = 0$ both numerator and denominator become zero. Let's use this fact, and apply l'Hôpital's Rule.

$$\begin{aligned} \lim_{x \rightarrow 0} \underbrace{\frac{e^{k \sin(x^2)} - (1 + 2x^2)}{x^4}}_{\substack{\text{num} \rightarrow 0 \\ \text{den} \rightarrow 0}} &= \lim_{x \rightarrow 0} \frac{2kx \cos(x^2) e^{k \sin(x^2)} - 4x}{4x^3} \\ &= \lim_{x \rightarrow 0} \frac{2k \cos(x^2) e^{k \sin(x^2)} - 4}{4x^2} \end{aligned}$$

When we plug in $x = 0$, the denominator becomes 0, and the numerator becomes $2k - 4$. So, we'll need some cases, because the behaviour of the limit depends on k .

For $k = 2$:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{2k \cos(x^2) e^{k \sin(x^2)} - 4}{4x^2} &= \lim_{x \rightarrow 0} \underbrace{\frac{4 \cos(x^2) e^{2 \sin(x^2)} - 4}{4x^2}}_{\substack{\text{num} \rightarrow 0 \\ \text{den} \rightarrow 0}} \\ &= \lim_{x \rightarrow 0} \frac{-8x \sin(x^2) e^{2 \sin(x^2)} + 16x \cos^2(x^2) e^{2 \sin(x^2)}}{8x} \\ &= \lim_{x \rightarrow 0} [-\sin(x^2) e^{2 \sin(x^2)} + 2 \cos^2(x^2) e^{2 \sin(x^2)}] \\ &= 2 \end{aligned}$$

For $k > 2$, the numerator goes to $2k - 4$, which is a positive constant, while the denominator goes to 0 from the right, so:

$$\lim_{x \rightarrow 0} \frac{2k \cos(x^2) e^{k \sin(x^2)} - 4}{4x^2} = \infty$$

For $k < 2$, the numerator goes to $2k - 4$, which is a negative constant, while the denominator goes to 0 from the right, so:

$$\lim_{x \rightarrow 0} \frac{2k \cos(x^2) e^{k \sin(x^2)} - 4}{4x^2} = -\infty$$

S-25:

- We want to find the limit as n goes to infinity of the percentage error,

$\lim_{n \rightarrow \infty} 100 \frac{|S(n) - A(n)|}{|S(n)|}$. Since $A(n)$ is a nicer function than $S(n)$, let's simplify:

$$\lim_{n \rightarrow \infty} 100 \frac{|S(n) - A(n)|}{|S(n)|} = 100 \left| 1 - \lim_{n \rightarrow \infty} \frac{A(n)}{S(n)} \right|.$$

We figure out this limit the natural way:

$$\begin{aligned} 100 \left| 1 - \lim_{n \rightarrow \infty} \frac{A(n)}{S(n)} \right| &= 100 \left| 1 - \lim_{n \rightarrow \infty} \frac{5n^4}{\underbrace{5n^4 - 13n^3 - 4n + \log(n)}_{\substack{\text{num} \rightarrow \infty \\ \text{den} \rightarrow \infty}}} \right| \\ &= 100 \left| 1 - \lim_{n \rightarrow \infty} \frac{20n^3}{20n^3 - 39n^2 - 4 + \frac{1}{n}} \right| \\ &= 100 \left| 1 - \lim_{n \rightarrow \infty} \frac{n^3}{n^3} \cdot \frac{20}{20 - \frac{39}{n} - \frac{4}{n^3} + \frac{1}{n^4}} \right| \\ &= 100 |1 - 1| = 0 \end{aligned}$$

So, as n gets larger and larger, the relative error in the approximation gets closer and closer to 0.

- Now, let's look at the absolute error.

$$\lim_{n \rightarrow \infty} |S(n) - A(n)| = \lim_{n \rightarrow \infty} |-13n^3 - 4n + \log n| = \infty$$

So although the error gets small *relative to the giant numbers we're talking about*, the absolute error grows without bound.

Solutions to Exercises 4.1 — Jump to [TABLE OF CONTENTS](#)

S-1: An antiderivative of $f'(x)$ is a function whose derivative is $f'(x)$. Our original function $f(x)$ has this property, so $f(x)$ is *an* antiderivative of $f'(x)$, but it's not the most general. We can add a constant to $f(x)$ without affecting its derivative. The most general antiderivative of $f'(x)$ is $f(x) + C$, where C is any constant.

S-2: Notice $f(x)$ is nonnegative for an interval covering the left part of the graph, and negative on the right part of the graph. That means its antiderivative is increasing for the left interval, then decreasing for the right interval. This applies to $A(x)$ and $C(x)$, but not $B(x)$.

There are only three points where $A(x)$ has a horizontal tangent line: at its global maximum and the endpoints of the interval shown. By contrast, $C(x)$ has a horizontal tangent line in four places: at its global maximum, at its inflection point, and at the endpoints of the interval shown. Since $f(x) = 0$ four times (and these line up with the horizontal portions of $C(x)$) we conclude $C(x)$ is the antiderivative of $f(x)$.

S-3: For any constant $n \neq -1$, an antiderivative of x^n is $\frac{1}{n+1}x^{n+1}$.

$$\begin{aligned} F'(x) &= 3x^2 + 5x^4 + 10x - 9 \\ F(x) &= 3 \left(\frac{1}{3} \right) x^3 + 5 \left(\frac{1}{5} \right) x^5 + 10 \left(\frac{1}{2} \right) x^2 - 9 \left(\frac{1}{1} \right) x^1 + C \\ &= x^3 + x^5 + 5x^2 - 9x + C \end{aligned}$$

Remark: we can always check by differentiating:

$$\begin{aligned} F'(x) &= \frac{d}{dx} \{x^3 + x^5 + 5x^2 - 9x + C\} \\ &= 3x^2 + 5x^4 + 10x - 9 \\ &= f(x) \end{aligned}$$

so $F(x)$ is indeed an antiderivative of $f(x)$.

S-4:

$$\begin{aligned} F'(x) &= \frac{3}{5}x^7 - 18x^4 + x \\ F(x) &= \left(\frac{3}{5} \right) \left(\frac{1}{8} \right) x^8 - 18 \left(\frac{1}{5} \right) x^5 + \frac{1}{2}x^2 + C \\ &= \frac{3}{40}x^8 - \frac{18}{5}x^5 + \frac{1}{2}x^2 + C \end{aligned}$$

S-5: For any constant $n \neq -1$, an antiderivative of x^n is $\frac{1}{n+1}x^{n+1}$. The constant n does not have to be an integer.

$$\begin{aligned} F'(x) &= 4\sqrt[3]{x} - \frac{9}{2x^{2.7}} \\ &= 4x^{\frac{1}{3}} - \frac{9}{2}x^{-2.7} \\ F(x) &= 4 \left(\frac{1}{\frac{1}{3}+1} \right) x^{\left(\frac{1}{3}+1\right)} - \left(\frac{9}{2} \right) \left(\frac{1}{-2.7+1} \right) x^{(-2.7+1)} + C \\ &= 4 \left(\frac{3}{4} \right) x^{\frac{4}{3}} - \left(\frac{9}{2} \right) \left(\frac{10}{-17} \right) x^{-1.7} + C \\ &= 3x^{\frac{4}{3}} + \frac{45}{17x^{1.7}} + C \end{aligned}$$

S-6:

- Solution 1: We can re-write $f(x)$ to make it a power of x .

$$\begin{aligned} F'(x) &= \frac{1}{7}x^{-\frac{1}{2}} \\ F(x) &= \left(\frac{1}{7}\right) \left(\frac{1}{-\frac{1}{2}+1}\right) x^{(-\frac{1}{2}+1)} + C \\ &= \left(\frac{1}{7}\right) (2)x^{\frac{1}{2}} + C \\ &= \frac{2}{7}\sqrt{x} + C \end{aligned}$$

- Solution 2: We notice that $\frac{1}{7\sqrt{x}}$ looks a lot like $\frac{1}{2\sqrt{x}}$, which is the derivative of $\sqrt{x}$. So:

$$\begin{aligned} \frac{d}{dx} \{\sqrt{x}\} &= \frac{1}{2\sqrt{x}} \\ \frac{d}{dx} \left\{ \frac{2}{7}\sqrt{x} \right\} &= \left(\frac{2}{7}\right) \frac{1}{2\sqrt{x}} = f(x) \end{aligned}$$

So, an antiderivative of $f(x)$ is $\frac{2}{7}\sqrt{x}$. Then the most general antiderivative is

$$F(x) = \frac{2}{7}\sqrt{x} + C.$$

S-7: We recall $\frac{d}{dx}e^x = e^x$. That is, e^x is its own antiderivative. So, a first guess for the antiderivative of $f(x)$ might be itself.

$$\frac{d}{dx} \{e^{5x+11}\} = 5e^{5x+11}$$

This isn't exactly right, so we modify it by multiplying by a constant.

$$\frac{d}{dx} \left\{ \frac{1}{5}e^{5x+11} \right\} = e^{5x+11}$$

This tells us that $\frac{1}{5}e^{5x+11}$ is an antiderivative of e^{5x+11} . Therefore, the most general antiderivative of e^{5x+11} is $F(x) = \frac{1}{5}e^{5x+11} + C$.

S-8: We know the derivatives of sine and cosine. We'll work from there to build a function whose derivative is $f(x)$. We'll start by finding an antiderivative of $7\cos(13x)$.

$$\begin{aligned} \frac{d}{dx} \{\sin x\} &= \cos x \\ \frac{d}{dx} \{\sin(13x)\} &= 13\cos(13x) \\ \frac{d}{dx} \left\{ \frac{1}{13}\sin(13x) \right\} &= \frac{13}{13}\cos(13x) = \cos(13x) \\ \frac{d}{dx} \left\{ \frac{7}{13}\sin(13x) \right\} &= 7\cos(13x) \end{aligned}$$

So, an antiderivative of $7 \cos(13x)$ is $\frac{7}{13} \sin(13x)$.

$$\begin{aligned}\frac{d}{dx}\{\cos x\} &= -\sin x \\ \frac{d}{dx}\{-\cos x\} &= \sin x \\ \frac{d}{dx}\{-\cos(5x)\} &= 5 \sin(5x) \\ \frac{d}{dx}\left\{-\frac{3}{5}\cos(5x)\right\} &= \left(\frac{3}{5}\right) 5 \sin(5x) = 3 \sin(5x)\end{aligned}$$

So, an antiderivative of $3 \sin(5x)$ is $-\frac{3}{5} \cos(5x)$.

The most general antiderivative of $3 \sin(5x) + 7 \cos(13x)$ is

$$F(x) = -\frac{3}{5} \cos(5x) + \frac{7}{13} \sin(13x) + C.$$

S-9: We know the derivative of $\tan x$ is $\sec^2 x$. Modifying this slightly, we see (using the chain rule)

$$\begin{aligned}\frac{d}{dx}\{\tan(x+1)\} &= \sec^2(x+1) \cdot \frac{d}{dx}\{x+1\} \\ &= \sec^2(x+1)\end{aligned}$$

So, $\tan(x+1)$ is an antiderivative of $\sec^2(x+1)$. Therefore, the most general antiderivative of $\sec^2(x+1)$ is $F(x) = \tan(x+1) + C$.

S-10: We note that $f(x)$ looks similar to $\frac{1}{x}$.

$$\begin{aligned}\frac{d}{dx}\{\log|x|\} &= \frac{1}{x} \\ \frac{d}{dx}\{\log|x+2|\} &= \frac{1}{x+2}\end{aligned}$$

The most general antiderivative of $f(x)$ is $F(x) = \log|x+2| + C$.

S-11: Our function $f(x)$ bears some resemblance to the derivative of arcsine, $\frac{1}{\sqrt{1-x^2}}$:

$$\begin{aligned}f(x) &= \frac{7}{\sqrt{3-3x^2}} = \frac{7}{\sqrt{3}} \left(\frac{1}{\sqrt{1-x^2}} \right) \\ \frac{d}{dx}\left\{\frac{7}{\sqrt{3}} \arcsin(x)\right\} &= \frac{7}{\sqrt{3}} \left(\frac{1}{\sqrt{1-x^2}} \right) = f(x)\end{aligned}$$

So, the most general antiderivative of $f(x)$ is $F(x) = \frac{7}{\sqrt{3}} \arcsin(x) + C$.

S-12: We notice that $f(x)$ looks similar to the derivative of the arctangent function, $\frac{1}{1+x^2}$.

$$f(x) = \frac{1}{1+25x^2} = \frac{1}{1+(5x)^2}$$

This gives us a first guess for our antiderivative: perhaps $\arctan(5x)$ will work. We test it by differentiating, making sure we don't forget the chain rule.

$$\frac{d}{dx} \{\arctan(5x)\} = \frac{1}{1 + (5x)^2} \cdot 5$$

We're close to $f(x)$, but we've multiplied by 5. That's easy to take care of: we can divide our guess by 5.

$$\begin{aligned} \frac{d}{dx} \left\{ \frac{1}{5} \arctan(5x) \right\} &= \frac{1}{5} \left(\frac{1}{1 + (5x)^2} \right) \cdot 5 \\ &= \frac{1}{1 + 25x^2} = f(x) \end{aligned}$$

So, the most general antiderivative of $f(x)$ is $F(x) = \frac{1}{5} \arctan(5x) + C$.

S-13: First, let's find the antiderivative of $f'(x)$. It's a polynomial, so we can use the observation from the text that an antiderivative of x^n , for any constant $n \neq -1$, is $\frac{1}{n+1}x^{n+1}$. Remember that the most general antiderivative will have an added constant.

$$\begin{aligned} f(x) &= \frac{3}{3}x^3 - \frac{9}{2}x^2 + 4x + C \\ &= x^3 - \frac{9}{2}x^2 + 4x + C \end{aligned}$$

Use the fact that $f(1) = 10$ to solve for C .

$$\begin{aligned} 10 &= 1 - \frac{9}{2} + 4 + C \\ C &= \frac{19}{2} \end{aligned}$$

All together,

$$f(x) = x^3 - \frac{9}{2}x^2 + 4x + \frac{19}{2}.$$

S-14: First, let's find the antiderivative of $f'(x)$. We know that one antiderivative of $\cos(x)$ is $\sin x$. We might guess that an antiderivative of $\cos(2x)$ is $\sin(2x)$. Check by differentiating:

$$\frac{d}{dx} \{\sin(2x)\} = 2 \cos(2x)$$

This is close to $f'(x)$, but we need to divide by 2.

$$\frac{d}{dx} \left\{ \frac{1}{2} \sin(2x) \right\} = \cos(2x)$$

So, $f(x) = \frac{1}{2} \sin(2x) + C$ for some constant C . We can find C using the given information $f(\pi) = \pi$.

$$\begin{aligned} \pi &= f(\pi) = \frac{1}{2} \sin(2\pi) + C \\ \pi &= C \end{aligned}$$

Therefore, $f(x) = \frac{1}{2} \sin(2x) + \pi$.

S-15: Looking at the table in the notes, we see the antiderivative of $\frac{1}{x}$ is $f(x) = \log|x| + C$. The given information $f(-1) = 0$ lets us find C :

$$\begin{aligned} 0 &= f(-1) = \log|-1| + C \\ 0 &= \log(1) + C \\ 0 &= C \end{aligned}$$

So, $f(x) = \log|x|$.

Remark: it is true that $\log x$ is an antiderivative of $\frac{1}{x}$, since the derivative of $\log x$ is $\frac{1}{x}$. However, $\log x$ is only defined for positive values of x . Since the desired $f(x)$ is defined for $x < 0$, you need to use a more general antiderivative of $\frac{1}{x}$: $\log|x| + C$.

S-16: An antiderivative of 1 is x , and an antiderivative of $\frac{1}{\sqrt{1-x^2}}$ is $\arcsin(x)$. So, $f(x) = \arcsin x + x + C$. The given information lets us find C .

$$\begin{aligned} -\frac{\pi}{2} &= f(1) = \arcsin(1) + 1 + C \\ -\frac{\pi}{2} &= \frac{\pi}{2} + 1 + C \\ C &= -\pi - 1 \end{aligned}$$

So, $f(x) = \arcsin x + x - \pi - 1$.

S-17: If $P(t)$ is the population at time t , then the information given in the problem is $P'(t) = 100e^{2t}$. Antidifferentiating, we see $P(t) = 50e^{2t} + C$, where C is some constant. We want to know for what value of t we get $P(t) = P(0) + 300$.

$$\begin{aligned} P(t) &= P(0) + 300 \\ 50e^{2t} + C &= 50e^0 + C + 300 \\ 50e^{2t} &= 350 \\ e^{2t} &= 7 \\ 2t &= \log(7) \\ t &= \frac{1}{2} \log(7) \end{aligned}$$

It takes $\frac{1}{2} \log 7$ hours (about 58 minutes) for the initial colony to increase by 300 individuals.

S-18: If $A(t)$ is the amount of money in your account at time t , then the given information is

$$A'(t) = 1500e^{\frac{t}{50}}$$

Antidifferentiating,

$$A(t) = 75000e^{\frac{t}{50}} + C$$

for some constant C .

That is, at time t , the amount of money in your bank account is $75000e^{\frac{t}{50}} + C$ dollars, for some constant C .

S-19: Let $P(t)$ be the amount of power your house has used since time $t = 0$. If t is measured in hours, and $P(t)$ in kWh (kilowatt-hours), then $P'(t)$ is the rate at which your house is consuming power, in kW. So, the given information is that

$$P'(t) = 0.5 \sin\left(\frac{\pi}{24}t\right) + 0.25$$

Antidifferentiating,

$$P(t) = -\frac{12}{\pi} \cos\left(\frac{\pi}{24}t\right) + 0.25t + C$$

Since $P(0)$ is the amount of energy consumed after 0 hours, $P(0) = 0$, so

$$\begin{aligned} 0 = P(0) &= -\frac{12}{\pi} + C \\ C &= \frac{12}{\pi} \\ P(t) &= \frac{12}{\pi} \left[1 - \cos\left(\frac{\pi}{24}t\right)\right] + 0.25t \end{aligned}$$

After 24 hours, your energy consumed is

$$\begin{aligned} P(24) &= \frac{12}{\pi} [1 - \cos(\pi)] + 0.25(24) \\ &= \frac{24}{\pi} + 6 \approx 13.6 \text{ kWh} \end{aligned}$$

S-20: We differentiate $f(x)$ and $g(x)$ using the chain rule. Recall $\frac{d}{dx} \{\arcsin(x)\} = \frac{1}{\sqrt{1-x^2}}$.

$$\begin{aligned} f'(x) &= \frac{2}{\sqrt{1-\sqrt{x}^2}} \cdot \frac{d}{dx} \sqrt{x} \\ &= \frac{2}{\sqrt{1-x}} \cdot \frac{1}{2\sqrt{x}} \\ &= \frac{1}{\sqrt{x-x^2}} \\ g'(x) &= \frac{1}{\sqrt{1-(2x-1)^2}} \cdot \frac{d}{dx} \{2x-1\} \\ &= \frac{2}{\sqrt{1-4x^2+4x-1}} \\ &= \frac{1}{\sqrt{x-x^2}} \end{aligned}$$

So, the derivative of $f(x) - g(x)$ is 0, which implies that $f(x)$ and $g(x)$ differ by a constant—perhaps a surprising result!

Remark: we don't need calculus to show that $f(x)$ and $g(x)$ only differ by a constant. Define $\theta = \sin^{-1} \sqrt{x}$, so that $\sin \theta = \sqrt{x}$ and $f(x) = 2\theta$, and then

$$\sin \left[f(x) - \frac{\pi}{2} \right] = -\cos f(x) = -\cos 2\theta = -[1 - 2\sin^2 \theta] = 2\sin^2 \theta - 1 = 2x - 1$$

Our goal is to take the arcsine of the first and last expressions and conclude

$f(x) - \frac{\pi}{2} = \arcsin(2x - 1) = g(x)$. However, before we can say

$\arcsin \left(\sin \left[f(x) - \frac{\pi}{2} \right] \right) = f(x) - \frac{\pi}{2}$, we have to check that $-\frac{\pi}{2} \leq f(x) - \frac{\pi}{2} \leq \frac{\pi}{2}$. That is, we need to show that $0 \leq f(x) \leq \pi$.

As $0 \leq \theta \leq \frac{\pi}{2}$ (since $\sin^{-1}$ always takes values between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$ and since $\sqrt{x} \geq 0$), we have that $0 \leq f(x) = 2\theta \leq \pi$. So,

$$f(x) - \frac{\pi}{2} = \arcsin \left(\sin \left[f(x) - \frac{\pi}{2} \right] \right) = \arcsin(2x - 1) = g(x).$$

S-21: The derivative of $\sin(2x)$ (which occurs in the second term) is $2 \cos(2x)$ (which occurs in the first term). Similarly, the derivative of $\cos(3x)$ (which occurs in the first term) is $-3 \sin(3x)$ (which occurs in the second term). So, $f(x)$ seems to have come from the product rule.

$$\frac{d}{dx} \{ \sin(2x) \cos(3x) \} = 2 \cos(2x) \cos(3x) - 3 \sin(2x) \sin(3x)$$

Then the antiderivative of $f(x)$ is $F(x) = \sin(2x) \cos(3x) + C$.

S-22: The function $f(x)$ looks like perhaps it came from the quotient rule. Recall

$$\frac{d}{dx} \left\{ \frac{u(x)}{v(x)} \right\} = \frac{v(x)u'(x) - u(x)v'(x)}{(v(x))^2}$$

Then, since the denominator of $f(x)$ is $(x^2 + 1)^2$, we might guess $v(x) = (x^2 + 1)$. That leaves $u(x) = e^x$.

$$\frac{d}{dx} \left\{ \frac{e^x}{x^2 + 1} \right\} = \frac{(x^2 + 1)e^x - e^x(2x)}{(x^2 + 1)^2} = f(x)$$

So, the antiderivative of $f(x)$ is $F(x) = \frac{e^x}{x^2 + 1} + C$.

S-23: We know that e^x is its own antiderivative. The derivative of $e^{g(x)}$, for some function $g(x)$, is $g'(x)e^{g(x)}$. Since $3x^2$ is the derivative of x^3 , $f(x)$ fits this pattern. We guess the antiderivative of $f(x)$ is $F(x) = e^{x^3} + C$.

We check by differentiating.

$$\frac{d}{dx} \{ e^{x^3} + C \} = 3x^2 e^{x^3} = f(x)$$

Indeed, the antiderivative of $f(x)$ is $F(x) = e^{x^3} + C$.

S-24: We know the antiderivative of $\sin x$ is $-\cos x$. Since $\sin(x^2)$ appears in our function, let's investigate the derivative of $-\cos(x^2)$.

$$\frac{d}{dx} \{-\cos(x^2)\} = \sin(x^2) \cdot 2x$$

This differs from $f(x)$ only by a constant multiple.

$$\frac{d}{dx} \left\{ -\frac{5}{2} \cos(x^2) \right\} = \frac{5}{2} \sin(x^2) \cdot 2x = 5x \sin(x^2) = f(x)$$

So, the antiderivative of $f(x)$ is $F(x) = -\frac{5}{2} \cos(x^2) + C$

Remark: as in Question **23**, our function $f(x)$ involved some function $g(x)$ as well as $g'(x)$. This pattern is the basis of an important method of antidifferentiation, called the Substitution Rule.

S-25: For any x in the domain of $\log(x)$, $e^{\log x} = x$. So, $f(x) = x$ for every x in its domain. Then its antiderivative is $F(x) = \frac{1}{2}x^2 + C$.

S-26: As in Question **11**, we notice that our function is similar to $\frac{1}{\sqrt{1-x^2}}$, but in this case it doesn't factor quite as nicely.

$$\begin{aligned} f(x) &= \frac{7}{\sqrt{3-x^2}} = \frac{7}{\sqrt{3(1-\frac{x^2}{3})}} \\ &= \frac{7}{\sqrt{3}} \left(\frac{1}{\sqrt{1-\frac{x^2}{3}}} \right) \end{aligned}$$

What we really want under that square root, instead of $\frac{x^2}{3}$, is simply x^2 . We can get close: we can get *something* squared.

$$f(x) = \frac{7}{\sqrt{3}} \left(\frac{1}{\sqrt{1-\left(\frac{x}{\sqrt{3}}\right)^2}} \right)$$

Now, the thing that's squared isn't x , it's $\frac{x}{\sqrt{3}}$. This gives us a first guess for an antiderivative:

perhaps $F(x) = \frac{7}{\sqrt{3}} \arcsin\left(\frac{x}{\sqrt{3}}\right)$ will work. Let's try it! Remember to use the chain rule when you differentiate.

$$\frac{d}{dx} \left\{ \frac{7}{\sqrt{3}} \arcsin\left(\frac{x}{\sqrt{3}}\right) \right\} = \frac{7}{\sqrt{3}} \left(\frac{1}{\sqrt{1-\left(\frac{x}{\sqrt{3}}\right)^2}} \right) \cdot \frac{1}{\sqrt{3}}$$

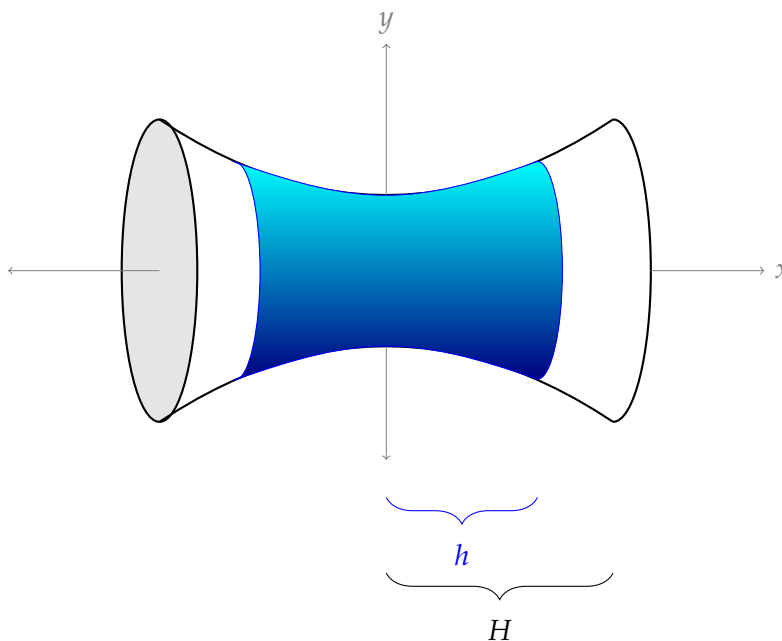
We're very close! We're only off by a constant, and those are easy to fix. We're dividing by $\sqrt{3}$ when we differentiate, so let's multiply our function by $\sqrt{3}$.

$$\begin{aligned}\frac{d}{dx} \left\{ 7 \arcsin \left(\frac{x}{\sqrt{3}} \right) \right\} &= 7 \left(\frac{1}{\sqrt{1 - \left(\frac{x}{\sqrt{3}} \right)^2}} \right) \cdot \frac{1}{\sqrt{3}} \\ &= \frac{7}{\sqrt{3} \sqrt{1 - \left(\frac{x}{\sqrt{3}} \right)^2}} \\ &= \frac{7}{\sqrt{3 - x^2}} = f(x)\end{aligned}$$

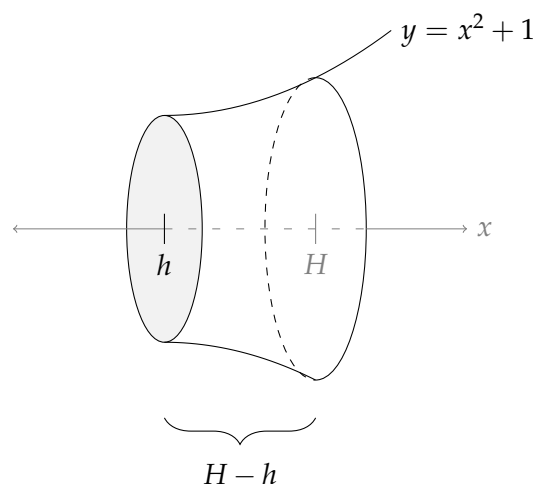
Therefore, the most general antiderivative of $f(x)$ is $F(x) = 7 \arcsin \left(\frac{x}{\sqrt{3}} \right) + C$.

S-27: Following Example 4.1.7, let $V(H)$ be the volume of the solid formed by rotating the segment of the parabola from $x = -H$ to $x = H$. **Our plan is to evaluate**
 $V'(H) = \lim_{h \rightarrow H} \frac{V(H) - V(h)}{H - h}$ **and then antidifferentiate $V'(H)$ to find $V(H)$. Since we don't know**
 $V(H) - V(h)$ **(yet) we first find upper and lower bounds on it when $h < H$.**

For a constant $h < H$, $V(H) - V(h)$ is the volume of the solid inside the larger object (with length $2H$) and outside the smaller object (with length $2h$, shown below in blue). There are two regions inside the larger object and outside the smaller.



One of the regions inside the larger object and outside the smaller (blue) object looks like this:



Remember we formed this solid by rotating the curve $y = x^2 + 1$ about the x -axis. So, the cross-section of this solid is a circle, and the radius of the circle when we are x units from the origin is $x^2 + 1$. So, the largest radius of the little shape shown above, which occurs at the right end, is $H^2 + 1$, and the smallest radius (at the left end) is $h^2 + 1$.

The volume of the shape shown above is less than a cylinder of radius $H^2 + 1$ and height $H - h$, and it is more than the volume of a cylinder of radius $h^2 + 1$ and height $H - h$. So, the volume of the shape shown above is between $(H - h)\pi(h^2 + 1)^2$ and $(H - h)\pi(H^2 + 1)^2$ cubic units.

Recall that the volume inside the object of length $2H$ and outside the object of length $2h$ consists of *two* copies of the shape shown above. Therefore:

$$2(H - h)\pi(h^2 + 1)^2 < V(H) - V(h) < 2(H - h)\pi(H^2 + 1)^2$$

Now that we have upper and lower bounds for $V(H) - V(h)$, we can find $V'(H)$.

$$\begin{aligned} \frac{2(H - h)\pi(h^2 + 1)^2}{H - h} &< \frac{V(H) - V(h)}{H - h} < \frac{2(H - h)\pi(H^2 + 1)^2}{H - h} \\ 2\pi(h^2 + 1)^2 &< \frac{V(H) - V(h)}{H - h} < 2\pi(H^2 + 1)^2 \\ \lim_{h \rightarrow H} 2\pi(h^2 + 1)^2 &\leq \lim_{h \rightarrow H} \frac{V(H) - V(h)}{H - h} \leq \lim_{h \rightarrow H} 2\pi(H^2 + 1)^2 \end{aligned}$$

Since the limits on both ends are simply $2\pi(H^2 + 1)^2$, by the Squeeze Theorem,

$$V'(H) = \lim_{h \rightarrow H} \frac{V(H) - V(h)}{H - h} = 2\pi(H^2 + 1)^2$$

Now that we know $V'(H)$, we antidifferentiate to find $V(H)$.

$$\begin{aligned} V'(H) &= 2\pi(H^4 + 2H^2 + 1) \\ V(H) &= 2\pi\left(\frac{1}{5}H^5 + \frac{2}{3}H^3 + H\right) + C \end{aligned}$$

When $H = 0$, there is no solid, so $V(0) = 0$. Therefore,

$$V(H) = 2\pi\left(\frac{1}{5}H^5 + \frac{2}{3}H^3 + H\right).$$

S-28: In Question **15**, we found that $f(x) = \log |x|$ for all $x < 0$. For all $x > 0$, $f(x) = \log x + C$, with C any constant, is the antiderivative of $1/x$. So

$$f(x) = \begin{cases} \log |x| & \text{for } x < 0 \\ \log x + C & \text{for } x > 0 \end{cases}$$

Remark: Note that $1/x$ and $\log |x|$ are both not defined for $x = 0$. If $\log |x|$ were defined at $x = 0$ and we also required that $f(x)$ be continuous at $x = 0$, C would be determined by the requirement that $\log |x|$ and $\log x + C$ take the same value at $x = 0$.